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WESTERN INTERNATIONAL SCHOOL

Physics



Grade 7

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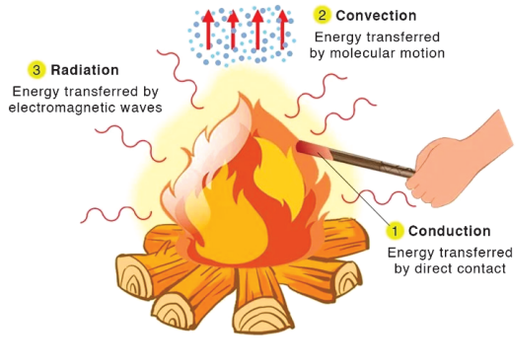
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Chapter 1 HEAT



LESSON 1



THERMOMETERS

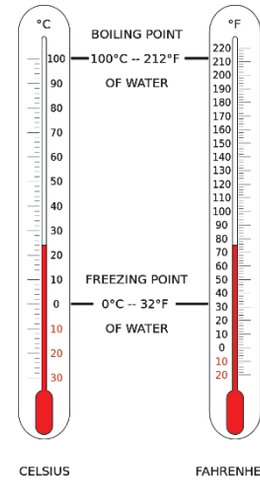
At the end of this chapter, you should be able to:

- Describe different types of thermometers
- Understand scale of thermometers
- Use clinical thermometer
- Convert temperature scales
- Differentiate heat and temperature.

The **temperature** of a body tells us how hot the body is. It is measured using a thermometer and usually in **degrees Celsius** (°C). There are different kinds of thermometer, each type being more suitable than another for a certain job. In each type the physical property used must vary continuously over a wide range of temperature. It must be accurately measurable with simple apparatus and vary in a similar way to other physical properties.

1. Liquid-in-glass thermometer

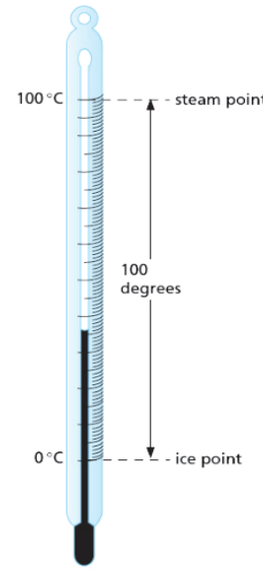
In this type the liquid in a glass bulb expands up a capillary tube when the bulb is heated. The liquid must be easily seen and must expand (or contract) rapidly and by a large amount over a wide range of temperature. It must not stick to the inside of the tube or the reading will be too high when the temperature is falling.



Mercury and colored alcohol are in common use. Mercury freezes at −39 °C and boils at 357 °C . Alcohol freezes at −115 °C and boils at 78 °C and is therefore more suitable for low temperatures.

2.Scales of Temperature

A scale and unit of temperature are obtained by choosing two temperatures, called the fixed points, and dividing the range between them into a number of equal divisions or degrees.

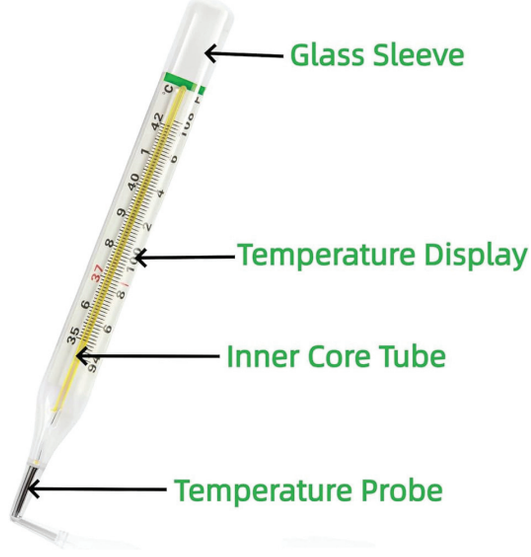


On the Celsius scale (named after the Swedish scientist who suggested it), the lower fixed point is the temperature of pure melting ice and is taken as 0 °C.

The upper fixed point is the temperature of the steam above water boiling at normal atmospheric pressure, 100 kPa (or N/m^2), and is taken as 100 °C. When the fixed points have been marked on the thermometer, the distance between them is divided into 100 equal degrees (Figure). The thermometer now has a linear scale, in other words it has been calibrated or graduated.

3. Clinical thermometer

A **clinical thermometer** is a special type of mercury-in-glass thermometer used by doctors and nurses. Its scale only extends over a few degrees on either side of the normal body temperature of 37 °C (Figure), i.e. it has a small **range**. Because of the very narrow capillary tube, temperatures can be measured very accurately, in other words, the thermometer has a high **sensitivity**. The tube has a constriction (a narrower part) just beyond the bulb. When the thermometer is placed under the tongue the mercury expands, forcing its way past the constriction. When the thermometer is removed (after 1 minute) from the mouth, the mercury in the bulb cools and contracts, breaking the mercury thread at the constriction. The mercury beyond the constriction stays in the tube and shows the body temperature. After use the mercury is returned to the bulb by a flick of the wrist. Since mercury is a toxic material, digital thermometers are now replacing mercury thermometers for clinical use.





4. Thermocouple thermometer

A thermocouple consists of wires of two different materials, such as copper and iron, joined together. When one junction is at a higher temperature than the other, an electric current flow and produces a reading on a sensitive meter which depends on the temperature difference.

Thermocouples are used in industry to measure a wide range of temperatures from −250 °C up to about 1500 °C, especially rapidly changing temperatures and those of small objects.

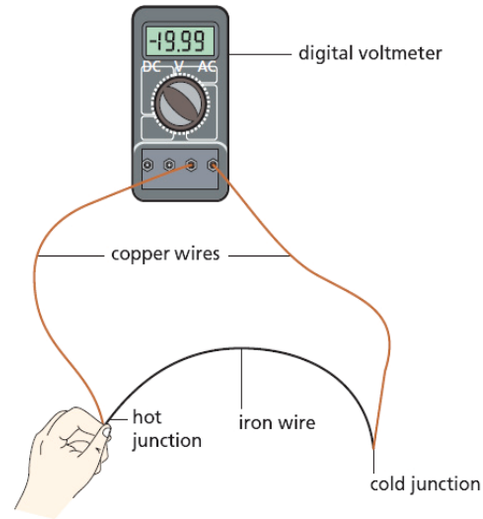


Figure 20.4 A simple thermocouple thermometer

5. Other thermometers

One type of **resistance thermometer** uses the fact that the electrical resistance of a platinum wire increases with temperature. A resistance thermometer can measure temperatures accurately in the range −200 °C to 1200 °C but it is bulky and best used for steady temperatures. A **thermistor** can also be used but over a small range, such as −5 °C to 70 °C; its resistance decreases with temperature.

The **constant-volume gas thermometer** uses the change in pressure of a gas to measure temperatures over a wide range. It is an accurate but bulky instrument. **Thermochromic liquids** which change color with temperature have a limited range around room temperatures.

6. Temperature Scale Conversion

a) To convert Celsius scale to Fahrenheit, we use the relationship:

$$T_F = \frac{9}{5} T_C + 32°$$

b) To convert Fahrenheit to Celsius scale, we use the relationship:

$$T_C = \frac{5}{9} (T_F - 32)$$

c) To convert Celsius scale to Kelvin, we use the relationship:

$$T_K = T_c + 273°$$

d) To convert Kelvin to Celsius scale, we use the relationship:

$$T_C = T_K - 273°$$

7. Heat and Temperature

It is important not to confuse the temperature of a body with the heat energy that can be obtained from it. For example, a red-hot spark from a fi re is at a higher temperature than the boiling water in a saucepan. In the boiling water the average kinetic energy. of the molecules is lower than in the spark; but since there are many more water molecules, their total energy is greater, and therefore more heat energy can be supplied by the water than by the spark. **Heat passes from a body at a higher temperature to one at a lower temperature**. This is because the average kinetic energy (and speed) of the molecules in the 'hot' body falls as a result of the collisions with molecules of the 'cold' body whose average kinetic energy, and therefore temperature, increases. When the average kinetic energy. of the molecules is the same in both bodies, they are at the same temperature. For example, if the red-hot spark landed in the boiling water, heat would pass from it to the water even though much more heat energy could be obtained from the water.

Heat is also called thermal or internal energy; it is the energy a body has because of the kinetic energy and the potential energy (p.e.) of its molecules. Increasing the temperature of a body increases its heat energy because the kinetic energy of its molecules increases. The internal energy of a body can also be increased by increasing the p.e. of its molecules.



End of Lesson

PROBLEMS

1. Convert the following to Fahrenheit
   1) 10º C _____
   2) 30º C _____
   3) 0º C _____
2. **Convert the following to Kelvin**
   1) 0º C _____
   2) -50º C _____
   3) 90º C _____
   4) -20º C _____
3. **Convert the following to Celsius**
   1) 112 K _____
   2) 200 K _____
   3) 273 K _____
   4) 350 K _____
4. **Conversion**

Convert the following to Fahrenheit	**Convert the following to Celsius**
1) 10º C _____	4) 32º F _____
2) 30º C _____	5) 90º F _____
3) 0º C _____	6) 212º F _____

Convert the following to Kelvin	**Convert the following to Celsius**
7) 0° C _____	11) 112° K _____
8) -50° C _____	12) 200° K _____
9) 90° C _____	13) 273° K _____
10) -20° C _____	14) 350° K _____



LESSON 2 QUANTITY OF HEAT

If 1 kg of water and 1 kg of paraffin are heated in turn for the same time by the same heater, the temperature rise of the paraffin is about twice that of the water. Since the heater gives equal amounts of heat energy to each liquid, it seems that different substances require different amounts of heat to cause the same temperature rise in the same mass, say 1ºC in 1kg. The 'thirst' of a substance for heat is measured by its specific heat capacity (symbol c).

The specific heat capacity of a substance is the heat required to produce a 1 ºC rise in 1 kg.

Heat, like other forms of energy, is measured in joules (J) and the unit of specific heat capacity is the joule per kilogram per ºC, i.e. J/(kg ºC).

In physics, the word 'specific' means that 'unit mass' is being considered.

1.Heat Equation

If a substance has a specific heat capacity of 1000 J/(kg ºC) then

- 1000 J raises the temperature of 1 kg by 1 ºC
- 2 x1000 J raises the temperature of 2 kg by 1 ºC
- 3 x 2 × 1000 J raises the temperature of 2 kg by 3 ºC

That is, 6000 J will raise the temperature of 2 kg of this substance by 3 ºC. We have obtained this answer by multiplying together:

(i) the mass in kg,

(ii) the temperature rises in ºC, and

(iii) the specific heat capacity in J/(kg ºC)

If the temperature of the substance fell by 3 °C, the heat given out would also be 6000 J. In general, we can write the heat equation as:
heat received or given out = mass × specific heat capacity ×temperature change.

In symbols: $\mathbf{Q = m \times c \times \Delta T}$

For example, if the temperature of a 5 kg mass of material of specific heat capacity 400 J/(kg ºC) rises from 15 ºC to 25 ºC, the heat received, Q , is Q = 5 kg × (25−15) ºC × 400 J/(kg ºC)

$$Q = 5\ kg \times 10\ ºC \times 400\ J/(kg\ ºC) = 20\ 000\ J$$

Table below shows specific heat capacity of some materials

Substance	Specific Heat Capacity J/kg°C	Substance	Specific Heat Capacity J/kg°C
Glass	670	Water	4200

Iron	460	copper	400
Brass	380	seawater	3900
Lead	130	Aluminum	900
Silver	240	Zinc	390

2.Thermal Capacity

The thermal capacity of a body is the quantity of heat needed to raise the temperature of the whole body by 1 °C. For a temperature rise of 1 °C the heat equation becomes:

heat received = mass × 1 × specific heat

capacity so that:

thermal capacity = mass × specific heat capacity = m × c

Thermal capacity is measured in joules per ºC, i.e. J/ºC.

For a copper block of mass 0.1 kg and specific heat capacity 390 J/(kg °C), thermal capacity = m × c = 0.1 kg × 390 J/(kg °C) = 39 J/°C

3.Practical work

Finding specific heat capacities, You need to know the power of the 12 V electric immersion heater to be used. (Precaution: Do not use one with a cracked seal.) A 40 W heater converts 40 joules of electrical energy into heat energy per second. If the power is not marked on the heater, ask about it.

a. Water

Weigh out 1 kg of water into a container, such as an aluminum saucepan. Note the temperature of the water, insert the heater (Figure 21.1), switch on the 12 V supply and start timing. Stir the water and after 5 minutes switch off, but continue stirring and note the highest temperature reached.

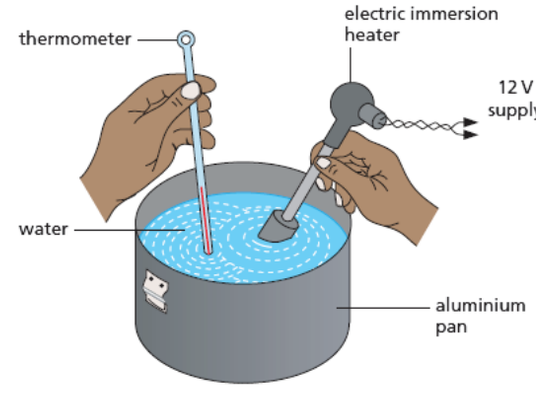


Figure 21.1

Figure 21.1 Assuming that the heat supplied by the heater equals the heat received by the water, work out the specific heat capacity of water in J/(kg ºC), as shown below: heat received by water (J) =power of heater (J/s) × time heater on(s).

Rearranging the 'heat equation' we get: specific heat capacity of water = heat received by water (J) / mass (kg) × temp. rise (ºC)

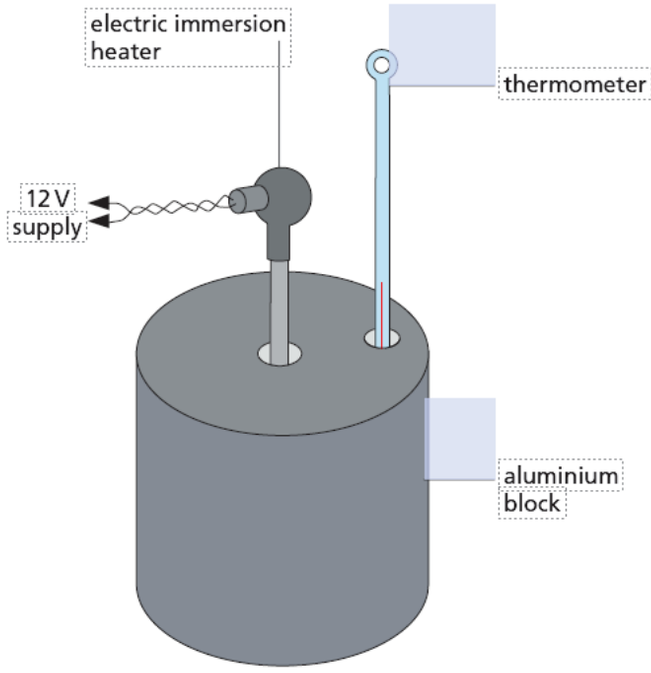


Figure 21.2

Suggest causes of error in this experiment.

b. Aluminum

An aluminum cylinder weighing 1 kg and having two holes drilled in it is used.
Place the immersion heater in the central hole and a thermometer in the other hole (Figure 21.2).
Note the temperature, connect the heater to a 12 V supply and switch it on for 5 minutes. When the temperature stops rising record its highest value.
Calculate the specific heat capacity as before.

4.Importance of the high specific heat capacity of water

The specific heat capacity of water is 4200 J/(kg ºC) and that of soil is about 800 J/(kg ºC). As a result, the temperature of the sea rises and falls more slowly than that of the land. A certain mass of water needs five times more heat than the same mass of soil for its temperature to rise by 1 ºC. Water also has to give out more heat to fall 1 ºC. Since islands are surrounded by water they experience much smaller changes of temperature from summer to winter than large land masses such as Central Asia.

The high specific heat capacity of water (as well as its cheapness and availability) accounts for its use in cooling engines and in the radiators of central heating systems.

Questions

1. How much heat is needed to raise the temperature by 10 ºC of 5 kg of a substance of specific heat capacity 300 J/(kg ºC)? What is the thermal capacity of the substance?
2. In kJ, how much heat is required to raise the temperature of 1.23 kg of water from 23.6 °C to 84.9 °C?
3. How much heat is released when a 260.0 g plate of copper cools from 5.30 x 10^2 K to 20.0 °C?
4. Find the quantity of heat required to raise the temperature of 500 g of iron by 2 °C, given that the Specific heat capacity is 500 J/(kg ^{0}C).



LESSON 3 Expansion of Solids, Liquids, and Gases



At the end of this chapter, you should be able to:

- Describe the use of liner expansion of solids,liquids,and gas
- Describe Precautions against expansion
- Explain Linear expansivity
- Understand the Unusual expansion of water

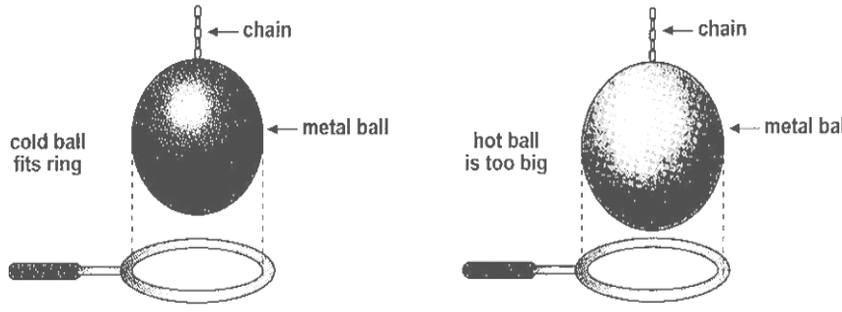
In general, when matter is heated it expands and when cooled it contracts. If the changes are resisted large forces are created which are sometimes useful but at other times are a nuisance.

According to the kinetic theory the molecules of solids and liquids are in constant vibration. When heated they vibrate faster and force each other a little further apart. Expansion results, and this is greater for liquids than for solids; gases expand even more. The linear (length) expansion of solids is small and for the effect to be noticed, the solid must be long and/or the temperature change must be large.

1.EXPANSION OF SOLIDS

Expansion is an increase in size of a substance. When heated, most substance increase in size in all directions.

Expansion of solids can be illustrated using a metal ball with a ring as shown below.



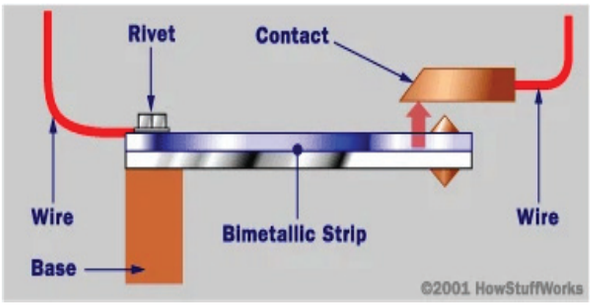
The metal ball passes through the ring when it is cold, but when heated, the ball doesn't pass through the ring any more, showing that it has expanded. It passes through the hole again when it cools, meaning that the metal contracts when it loses heat.

Different metals expand at different rates when equally heated, this can be shown using a metal strip made of two metals such as copper and iron bounded tightly together (bi- metallic strip).

APPLICATION OF EXPANSION OF SOLIDS

A.Bimetallic Strip

If equal lengths of two different metals, such as copper and iron, are riveted together so that they cannot move separately, they form a bimetallic strip (Figure a). When heated, copper expands more than iron and to allow this the strip bends with copper on the outside (Figure b). If they had expanded equally, the strip would have stayed straight.



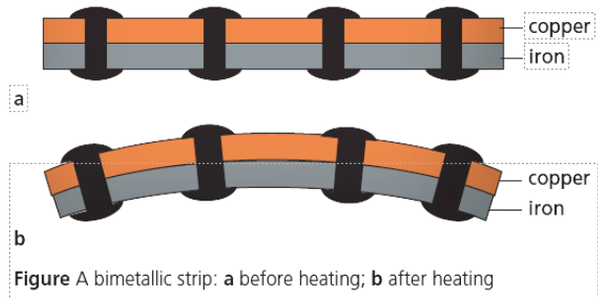
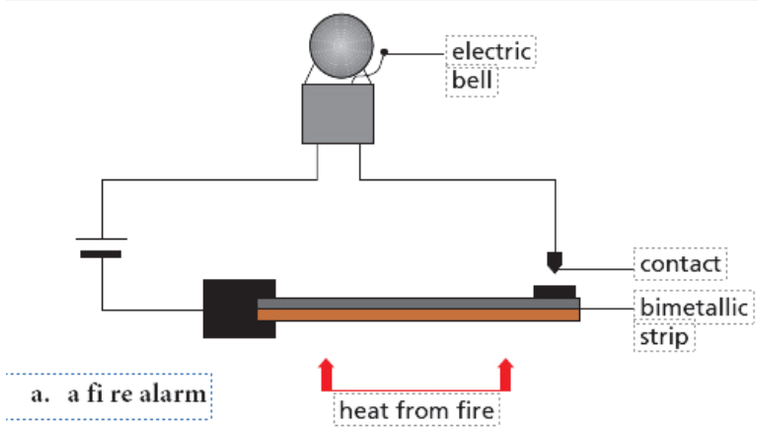


Figure A bimetallic strip: **a** before heating; **b** after heating

a. Fire alarm

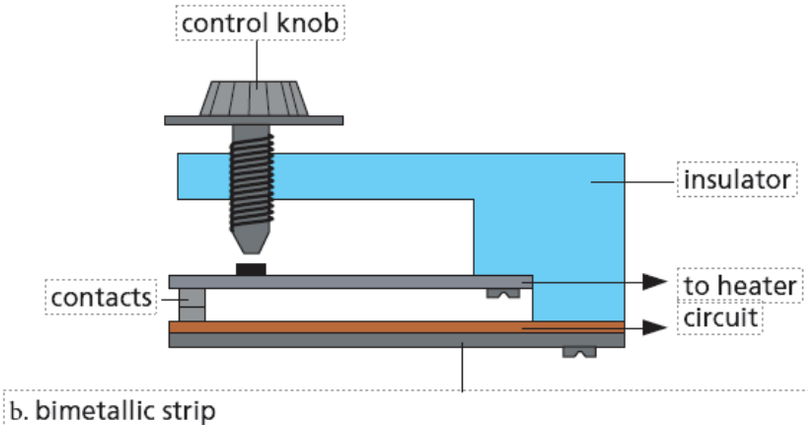
Heat from the fire makes the bimetallic strip bend and complete the electrical circuit, so ringing the alarm bell.

A bimetallic strip is also used in this way to work. the flashing direction indicator lamps in a car, being warmed by an electric heating coil wound round it.



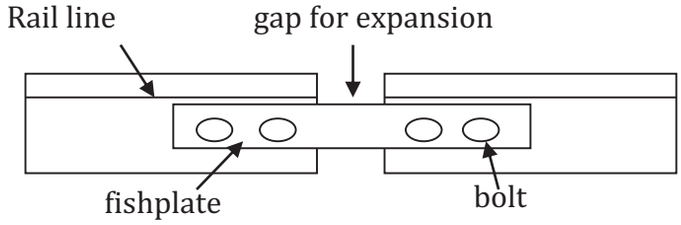
b. Thermostat

A thermostat keeps the temperature of a room or an appliance constant. The one in Figure b uses a bimetallic strip in the electrical heating circuit of, for example, an iron. When the iron reaches the required temperature the strip bends down, breaks the circuit at the contacts and switches off the heater. After cooling a little the strip remakes contact and turns the heater on again. A near-steady temperature results. If the control knob is screwed down, the strip has to bend more to break the heating circuit and this needs a higher temperature.

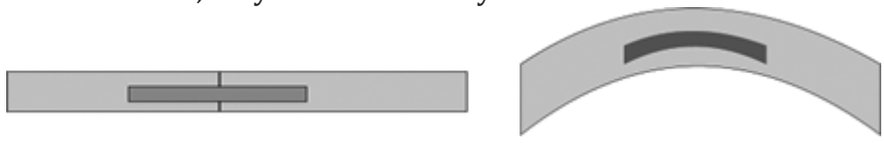


DISADVANTAGES OF EXPANSION Expansion can cause a number of problems:

- Contraction of railway lines, bridges, oil pipes and putting up electrical transmission wires. Therefore, allowance must be made for expansion.
- **Railway lines are** constructed with gaps left in between consecutive rails such that on hot days, when the rails expand; they have enough room for expansion.



If no gap is left in the rails, they bend on hot days.



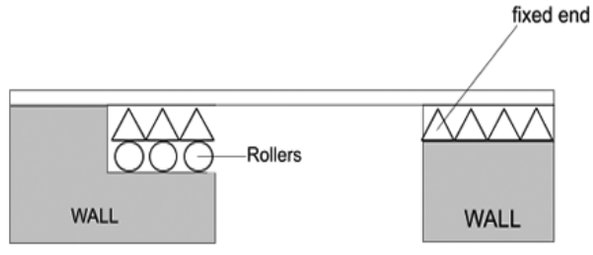
Modern rail lines are constructed so that they overlap as shown below.



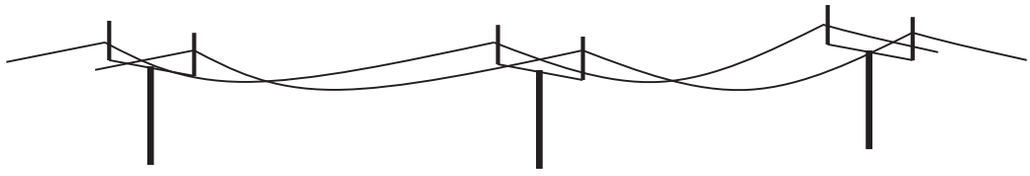
During expansion, the rails simply slide along each other.



- **Steam pipes and telephone wires:**
  Steam pipes are built to have flexible expansion joints in them to allow room for expansion.
- **Steel bridges**
  These are constructed in such a way that one end is rested on rollers and the other end is normally fixed. This is to ensure that the structure can contract and expand freely at various temperatures without damaging the bridge.

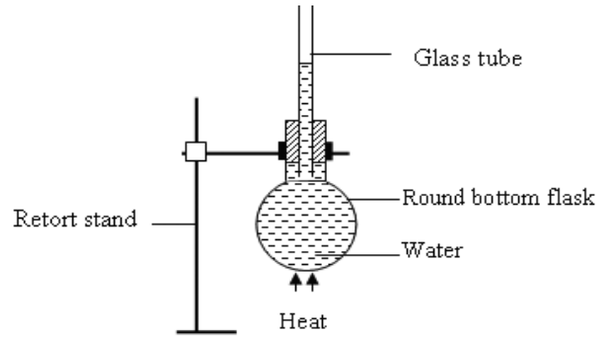


- **Transmission cables**
  Wires or cables in transmission or telephone cables are normally not pulled tightly during installation but are loosely held or have loops at various intervals over long distances in order to allow room for expansion and contraction during extreme weather conditions.



2.EXPANSION of LIQUIDS

When liquids or gases (fluids) get hot, they expand just as solids do, but their expansion is greater than that of solids for the same amount of heat. Since liquids do not have fixed length or surface area, but always take up the shape of the vessel, it's their volume changes that are considered when they are heated.



Unusual Expansion of Water

As water is cooled to 4 ºC it contracts, as we would expect. However, between 4 ºC and 0 ºC it expands, surprisingly. Water has a maximum density at 4 ºC (Figure 19.5).

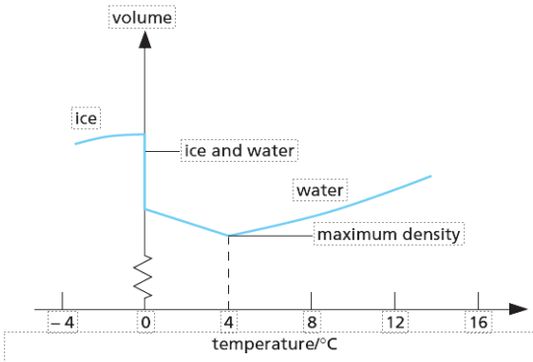


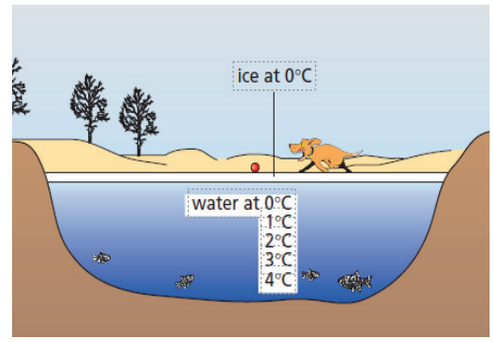
Figure 19.5 Water expands on cooling below 4 °C.



Figure 19.6 Result of the expansion of water on freezing

At 0 ºC, when it freezes, a considerable volume expansion occurs and every 100 cm^3 of water becomes 109 cm^3 of ice. This accounts for the bursting of unlagged water pipes in very cold weather and for the fact that ice is less dense than cold water and so floats. Figure 19.6 shows a bottle of frozen milk, the main constituent of which is water.

The unusual expansion of water between 4ºC and 0ºC explains why fish survive in a frozen pond. The water at the top of the pond cools first, contracts and being denser sinks to the bottom. Warmer, less dense water rises to the surface to be cooled. When all the water is at 4ºC the circulation stops. If the temperature of the surface water falls below 4ºC, it becomes less dense and remains at the top, eventually forming a layer of ice at 0ºC. Temperatures in the pond are then as in Figure.

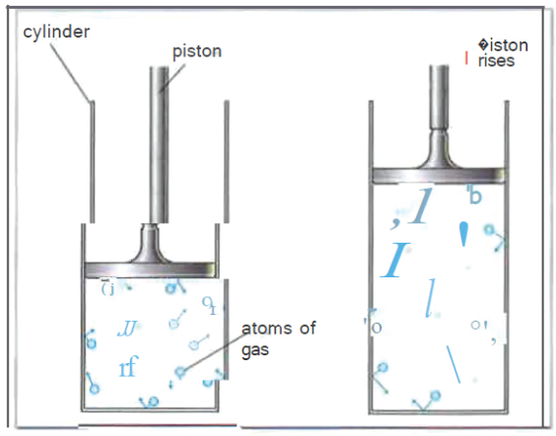


The volume expansion of water between 4ºC and 0ºC is due to the breaking up of the groups that water molecules form above 4ºC. The new arrangement requires a larger volume and more than cancels out the contraction due to the fall in temperature.

3.Expansion of Gas

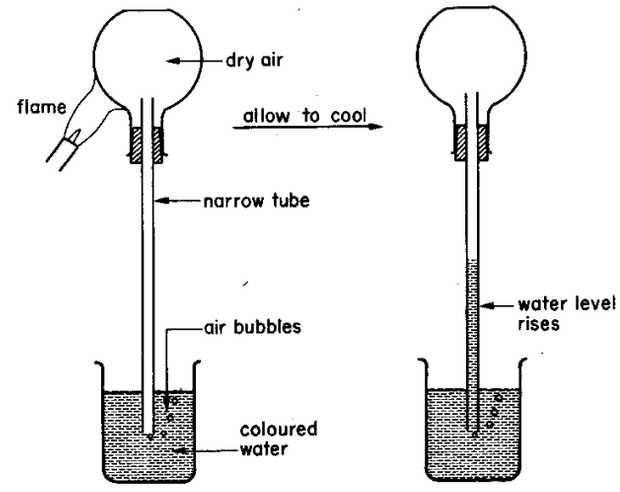
Gases expand when they are heated, just like solids and liquids. We can understand this using the kinetic model of matter. Figure shows some gas in a cylinder fitted with a piston. At first, the gas is cold and its particles press weakly on the piston. When the gas is heated, its particles move faster. Now they push with greater force on the piston and push it upwards. The gas has expanded.

The upward force of the gas is balanced by the downward weight of the piston. So, in this situation, the pressure of the gas has remained constant as it has expanded. If the piston did not move, the *volume* of the gas would remain constant when it was heated but its *pressure* would increase.



EXPERIMENT TO DEMONSTRATE EXPANSION IN GASES

- Coloured water is poured in a beaker.
- A flat bottomed flask is obtained and is tightly closed with a rubber cork with narrow tube fixed through it.
- The flask is slightly heated.
- Air bubbles will be seen coming out from the other end of the tube.
- This shows that air expands when heated.
- In the second set up, when the surface of heat is removed and the flask is allowed to cool by pouring cold water, the level of water will rise. This shows that air contacts when cooled.



Application of expansion of air.

Hot air balloon

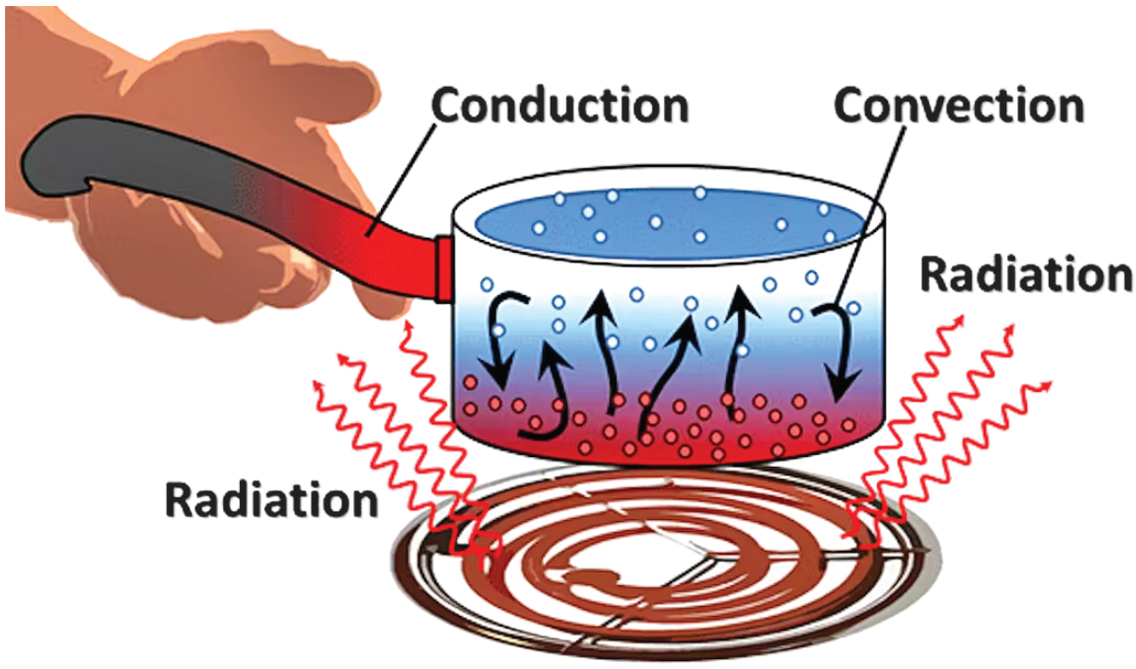
Expansion of air is used in hot air balloons. When air in the balloon is heated, it expands and becomes less dense and as a result the balloon rises up.



LESSON 4 Heat Transfer

Objectives: After completing this unit you should be able to:

- ✓ understand concepts related to heat energy.
- ✓ develop skill of manipulating problems related to heat energy.
- ✓ appreciate the interrelatedness of all things.
- ✓ use a wide range of possibilities for developing knowledge of the major concepts with in physics.



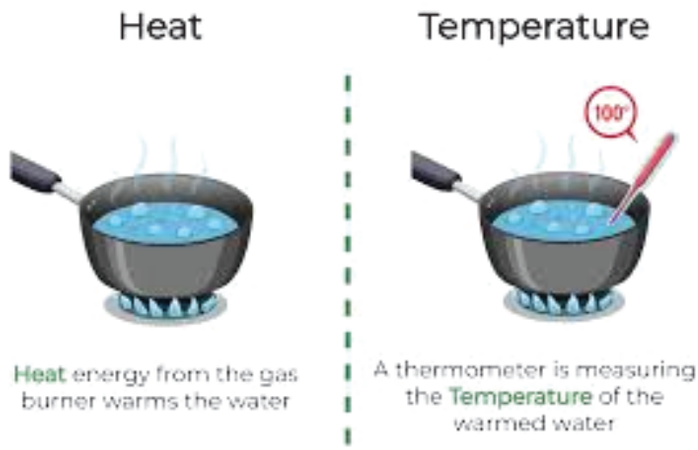
1.Introduction

In Lesson 1, you learnt about temperature and heat. You learnt the differences between temperature and heat. Further more you studied about temperature scales, measuring temperature, types of thermometers and some effects of heating a body. In this grade and this unit you will learn about heat transferring methods and the quantity of heat energy gained or lost by a substance.

Under heat transfer you will learn the three basic mechanisms of heat transfer namely conduction, convection and radiation and be introduced to different practical activities which help you to understand these heat transfer mechanisms. The diagram on page 68 shows the different ways bodies get heat from a burning wood. Can you explain the different ways shown in the diagram?

When you deal with quantity of heat you will learn how to calculate heat lost or gained by a body. The relation between quantity of heat and change in temperature and nature of the substance will be your main focus. You will also be introduced to the concept of "Specific heat capacity".

Activity 1. Discuss with your friends on the following points.
i) What is temperature? And how is it related to heat? ii) Explain the interrelationship between heat and molecular energy of a body. iii) Is heat transferable? How do bodies get heat from different source?



Temperature is the measure of hotness or coldness of a body. It is the measure of the average molecular kinetic energy.
It is an indication of the direction of heat flow.
Kelvin is the SI uint of temperature.
Thermometer is the instrument used to measure temperature

Can you,explain them?The term temperatureis one of the fundamental physical quantities. It is the measure of hotness or coldness of a body. Heat is a form of energy. Heat energy is directly related with the kinetic energy of molecules of a body. The kinetic energy of the atoms and molecules of matter gives rise to heat energy.

Heat transfer is the movement or flow of heat energy from a hotter body to a colder one.

Heat transfer occurs whenever there is a temperature difference between two bodies or between two parts of the same body. The movement of heat stops when the temperature of each object becomes the same. There are three ways by which heat moves from one body to another. These are Conduction, Convection and Radiation.

2. Conduction

Activity 2: Observing the transfer of heat by conduction

Material required: **A metal rod, a source of heat (candle) wax, and paper pins.**

Procedures: **Fix the paper pins into wax around the metal rod.**

i. **Hold one end of the metal rod, and place its other end into a burning candle.(As shown in Fig. 1)**
ii. **After a period of time, the wax near the candle melts and the pin falls.**
iii. **How did the heat move from the source to the metal rod?**
iv. **What do we call this method of heat transfer?**

Heat energy can be transferred by conduction from one substance to another when they are in direct contact.

Conduction is one way of heat transfer. It takes place from a hot body to a cold body by neighboring particles.

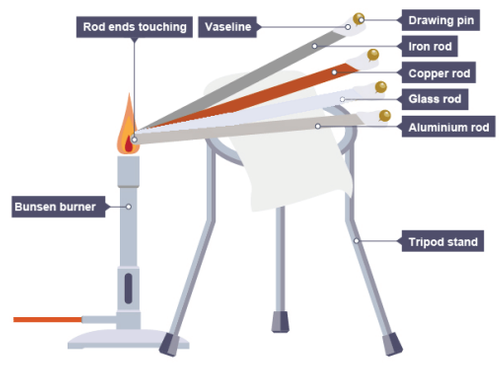


Fig 1. Conduction of heat in a metal

Conduction is mainly seen with in solid objects; however it can happen when any material comes into contact with a source of heat. When you hold a cup of hot water, your hand gets heat by conduction from the hot water.



Activity 3: Classifying materials as conductors and insulators of heat

Materials required: The same materials as in Activity 2, and different materials like copper, steel, aluminum, wood, glass, plastic, etc.

Procedure: Use the procedure indicated in Activity 2. for different materials. Fill in the table below

Materials	Good conductor	Poor conductor (insulator)
Copper		
Steel		
Wood		
Glass		

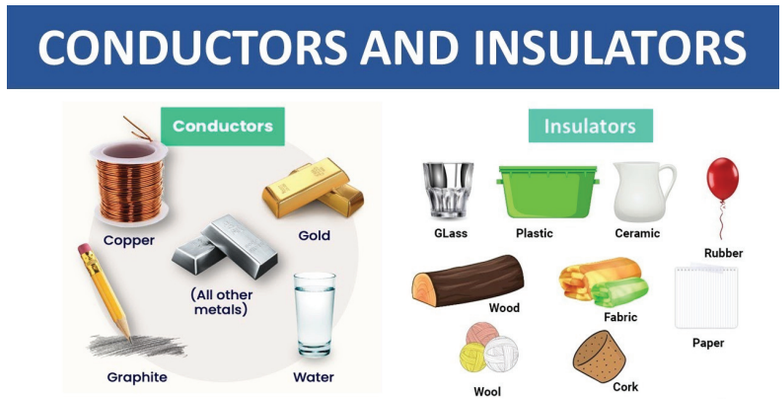
The degree with which heat flows within matter varies greatly.

Some materials are better conductors of heat than others. For example, metals are good conductors of heat, while a material like wood isn't. Metal heated on one end will soon be hot on the other end too, while this is not true with a piece of wood. Good conductors of electricity are often good conductors of heat.

Conduction is good in solid and it is poor within gases, because the atoms and molecules are not in close contact with each other.

Since the atoms in solid are closer to each other, solids conduct heat better than liquids or gasses. This means that two solid materials in contact would transfer heat from one to the other better than a solid in contact with a liquid.

Materials which allow heat to flow through them are called heat conductors. Iron, copper and other metals are examples of heat conductors.
Materials which do not allow heat to pass through them are called heat *Insulators*. Wool, wood, water glass, etc are examples of heat insulators.

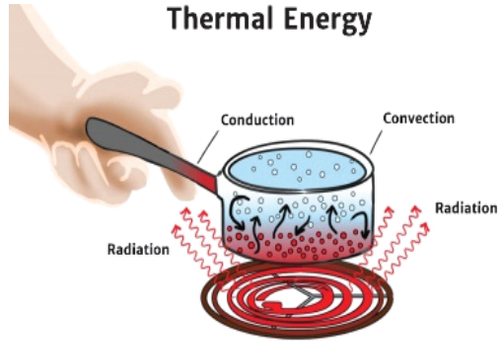


What happens in conduction?

Heat flows from a hotter part of a body to colder part of a body. During heating, the molecules at the hot end of the material increase their vibrations since the temperature on the region is increased. As a result they collide with their slowly moving neighbors and their kinetic energy is transmitted successively from one particle to the next, until it reaches the colder region. Hence heat energy is transmitted from one end to the other while the particles are fixed in their original position except for their violent vibration about their original position.

Methods of controlling heat loss

Heat conduction is important in our daily lives. In cold weather, we wear clothes that stop the conduction of heat from our warm bodies to the cold air. For example, we wear head cap, hand gloves, cotton jacket, and wear socks to keep heat in our body.



Activity 4: Practicalwork

i. **Put a pan on a hot stove. Can you touch the pan after some seconds? Why?**
ii. **Why do you think the handle of a pan is made of wood or plastic?**
iii. **Explain why people use carpets and rugs in their room.**
iv. **Stand on your bare feet on a ceramic tiled floor. What do you feel?**



From Activity 4 and Fig. 2 you will notice that, when a pan is placed on hot stove, the bottom of the pan is in contact with a hot burner. The heat flows through the metal and soon raises the temperature of all parts of the pan.

Similarly, when you step on a carpets very little heat is transferred away from your feet, so you stay warm. But when you step bare feet on cement floor or ceramic tiled floor, more heat is transferred from your feet. So you feel cold.

Handles on metal pan are often made of wood or plastics to stop the conduction of heat from the burner to your hand as shown in Fig 3.

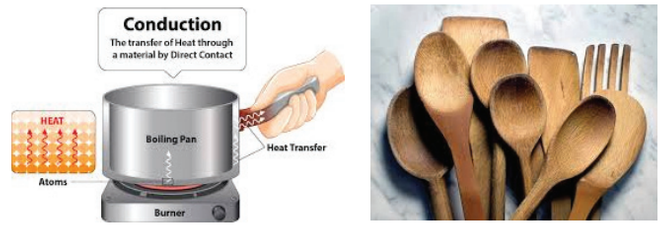


Fig 3. Examples of good conductors and poor conductors

Carpets and rugs are also used to reduce loss of heat by conduction.

Activity 5
How do your families keep heat in foods (drinks) at home? Discuss this question with your parents.

3. Convection

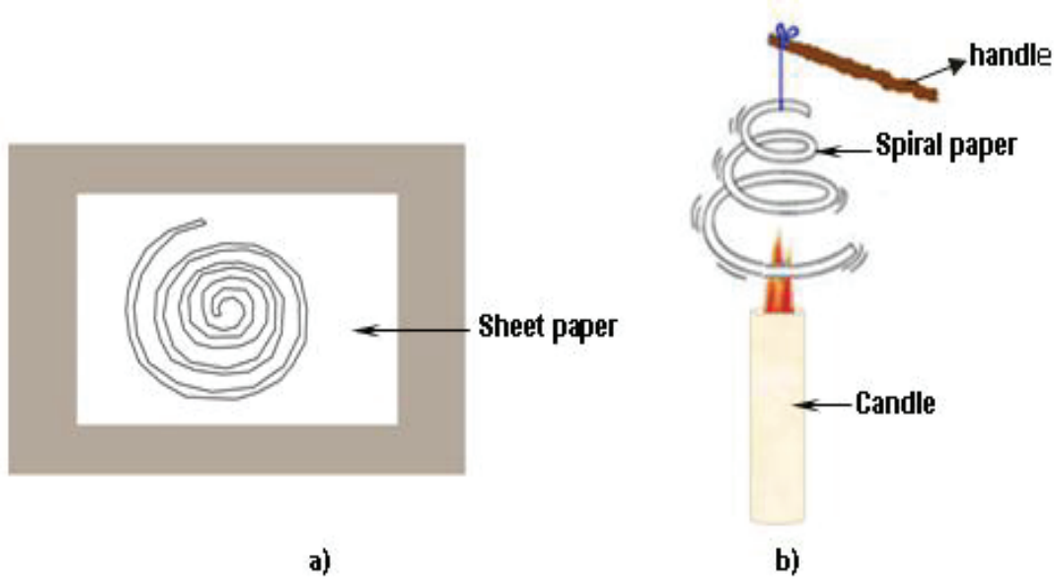


Fig 4.Observing the effect of heat transfer by convection

Activity 6: Observing heated air rises
Material: **A sheet of paper, scissors, and thread.** *Procedure:* Cut the paper along the line spirally. (Fig. 4.(a)) i. Hang the cut spiral paper by passing a string (thread) along it center. ii. Put a burning candle under heating the spiral paper.(Fig. 4.(b)) **iii.** Observe and explain what happens to the paper. **iv.** Explain what causes it to rotate.

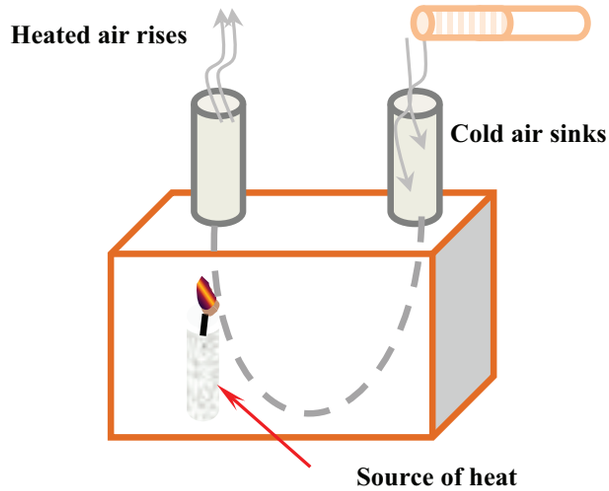


Fig 5 . Convection in air (Smoke box)

Activity 7. Observing convection in air

Material required: **A cardboard box with one side covered with transparent plastic sheet.**

i. **A source of heat**
ii. **Two tubes made from paper sheet.**
iii. **A smoking paper**

Procedure: **Assemble the materials as shown in Fig 5. Observe and explain what happens to the smoke.**

- Explain the process of convection in this demonstration.

In Fig 4. and 5, the air gets heated by the burning candle. The heated air rises because it expands and become less dense than the air around it. Cooler air, which is heavier, goes down and replaces the hot air place. This cold air in turn is heated. Fig 4. and Fig 5 shows transfer of heat by convection in air.

The transfer of heat from one place to another by the actual movement of particles of the medium is called convection. Convection takes place primarily in fluids (gas and liquids). It is caused by the expansion of fluid particles when it is heated. As fluid particles expand, its density decreases. Thus the lighter portion of the fluid rises, and cold fluids takes its place and get heated. In convection, heat is carried by the moving medium. Convection can occur only in liquids and gases.

Transfer of heat by convection is applied in home, industry and in nature.

1. Air conditioner, chimney and boiling water in a dish use convection.
2. Land and sea breezes also work with convection. Wind blows from sea to land during the night, while wind blows from land to sea during the day because of temperature difference
3. The balloon shown in Fig 6 uses convection. A hot air balloon contains heated air having less density than the surrounding air. Thus the heated air rises up taking the balloon with it. For what purpose is a hot air balloon used?



Fig 6. Uses of convection in a balloon

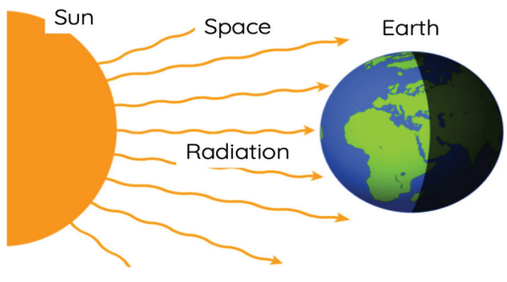
4. Radiation

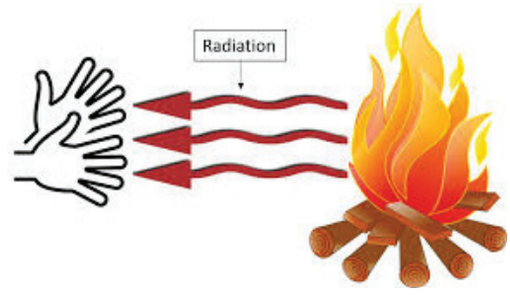
Activity 8

***Discuss*: i) How does the heat from the sun reach our earth?**
ii) How does the sunlight change into heat on the earth's surface?
iii) What do you call this type of heat transfer?

Heat transferred by radiation is also called **heat radiation**. Heat radiation is carried by an electromagnetic wave called **infra-red wave**. This wave is not visible to human eyes.

All objects emit heat radiation. For example, if an object is hotter than its surroundings, then it will give out heat radiation. In doing so, it will lose heat and cool down. Heat from electric heater and heat from a burning charcoal reaches its surrounding by radiation. For example, when we sit around a fire we feel warm. Conduction or convection is not responsible here. Since air is a poor conductor of heat and the warm air rises upwards. The heat reaches to our body from the fire by radiation. Heat can be transmitted by radiation through vacuum or a material medium.





Radiation **is the transfer of heat from one place to another without a material medium. The heat of the sun reaches the earth by the process of radiation. Heat can be transferred by radiation in vacuum.**

Types of surfaces will affect how quickly the object heats up. A rough and black surface is a good emitter and absorber of heat radiation while shiny and polished flat surface is a poor emitter and absorber of heat radiation.

Activity 9: Discuss the following with your friends.

- **What do you know of the terms: absorber, reflector and emitter of heat? Describe them and give examples related to heat absorption, reflection and emission.**

Activity 10: To observe which materials are good absorbers of heat radiation.

Material required: **Two identical cans; one painted white and the other painted black, thermometer, the sunlight.**

Procedure: **Fill the two cans with equal volume of water**

- **Put a thermometer in each can and leave both cans in the sunlight.**
- **Measure the temperature of water in both cans, and record the reading data at an interval of time, as shown in Fig 7.**

-Draw the graph of temperature versus time taken for both cans.

-Which material (can) absorbed more heat radiation?

Heat radiation is controlled in house construction, in types of clothes worn, and thermos flask. For example, houses built in very cold climate use special materials for flooring, walls and roofing. The heat supplied to warm the house is absorbed by good absorbers and emitted back to warm the room (house). People living in hot areas wear white clothes to reflect the heat radiation because white clothes are poor heat absorbers.

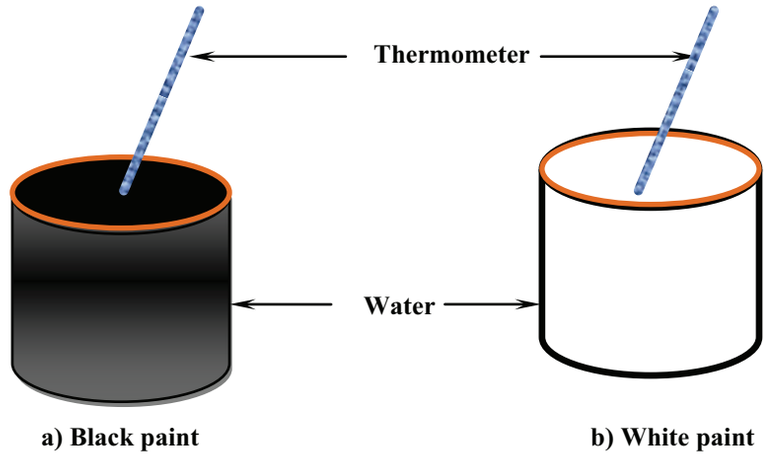


Fig 7 Observing good absorbers of heat radiation

Chapter 2 Electricity



Lesson 1 Static Electricity



When two materials rub together, there is friction. This friction causes rubbed materials to attract unrubbed materials. For example, if a rod of glass is rubbed with fur, it gains the power to attract light bodies such as pieces of paper or aluminum-foil or a pith ball. A body made attractive by rubbing is said to be 'electrified' or 'charged'.

The branch of electricity that studies electric charges at rest is called Electrostatics.

1.What are electric charges?

Activity 1
Answer the following questions with your friends. **i. What is an atom?** **ii. What are the two types of charges?** **iii. What type of charge is carried by protons?** **iv. What type of charge is carried by electrons?** **v. When do we say an atom is electrically charged?**

The results of numerous experiments indicated that under certain circumstances, matter exhibited a 'new' property. This property was eventually traced to the atom and is called **electric charge**.

Electric charge is an inherent physical property of certain sub-atomic particles that is responsible for electrical and magnetic phenomena. Charge is represented by 'Q' and the unit of charge is the Coulomb (C).

The smallest particle of an element is called an **atom**. The three important particles of an atom: **protons**, **electrons** and **neutrons** are known as the basic building blocks of all atoms. Fig 8.10 shows a model of how these particles are thought to form an atom. The protons and the neutrons are held together at the centrer of the atom called the **nucleus**. The electrons exist around the nucleus. They revolve in orbits around the nucleus at high speed.

Electric charge is a characteristic property of many subatomic particles. Electric charge is the property of a matter that exhibits attraction or repulsion over other matter.

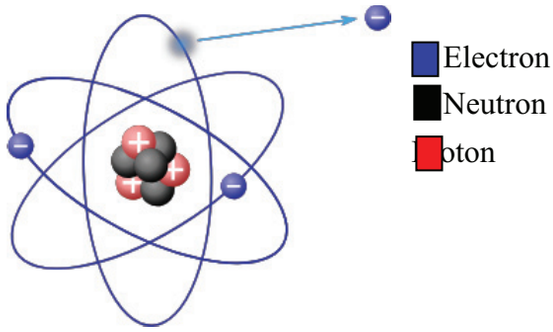


Fig 1.Structure of an atom

Types of charges

The idea of electricity is based on the theory that all matter are built up of atoms. Each atom has a tiny core called the nucleus and around this nucleus spins a number of electrons. The nucleus consists of neutron and proton.

Negative and positive charges

As it is mentioned above an atom consists of neutron, proton and electron particles.

- Neutron is a neutral particle; it does not have a charge.
- Proton is positively charged particle.
- Electron is a negatively charged particle.

Hence there are two types of electric charges called positive and negative charges. A negative charge is carried by electrons and a positive charge is carried by protons.

The allocation of signs to charges was made randomly many years ago. We use this assignment today.

Note the following points carefully:

i. Every matter is electrical in nature.

ii. We have two types of charges.

iii. In the normal state, the number of electrons is equal to the number of proton in an atom. Therefore, an atom is neutral as a whole. This explains why a body does not exhibit any charge under ordinary conditions.

iv. If some electrons are removed from a neutral body, there occurs a deficit of electrons in the body and consequently the body has surplus positive charge.

v. If a neutral body is supplied with electrons, there occurs an excess of electron and consequently the body is said to have a negative charge.

Check points

1. What does the term electrostatics stand for?
2. Describe the two types of electric charges.
3. What charges are carried by
   a) Protons
   b) Electrons

2. Methods of Charging a Body

Charging is the process of electrifying bodies i.e. removing from or adding electrons to a body.

There are different methods of charging a body:-

i. Charging by rubb **ii. Charging by conduction**

i.Charging by rubbing

Activity 2 Charging a body by rubbing

***Apparatus*:- a piece of paper, a polythene rod (plastic comb, plastic ball point, pen.) woolen cloth or you can use your hair.**

***Procedure*:**

i. **Tear pieces of paper into bits and put them on the table.**
ii. **Touch the bits of paper with a polythene rod without rubbing, observe what happens.**
iii. **Now rub the polythene rod with woolen cloths (your hair) and bring it close to the bits of paper. Observe what happens. Do you explain why it happens this way?**



Fig 2 Charged plastic rod attracts pieces of paper

From the structure of an atom we could understand that the rubbing of two bodies cause a transfer of electrons from one body to another. One of the bodies gains electrons, while the other loses electrons.

For example in Activity 2, when a plastic rod is rubbed with woolen clothes (your hair) electrons will be transferred from the plastic rod to the woolen cloth (you hair). Hence, the plastic rod is negatively charged while the woolen clothes is charged positively.
Similarly when a glass rod is rubbed with silk, electrons will be transferred from the glass to the silk. Hence the glass is positively charged while the silk is negatively charged.

ii. Charging by conduction (sharing)

When a neutral body is made to contact or touch a charged body, it shares the charges and becomes charged. Charging by conduction involves sharing of electrons between the charged and uncharged bodies.

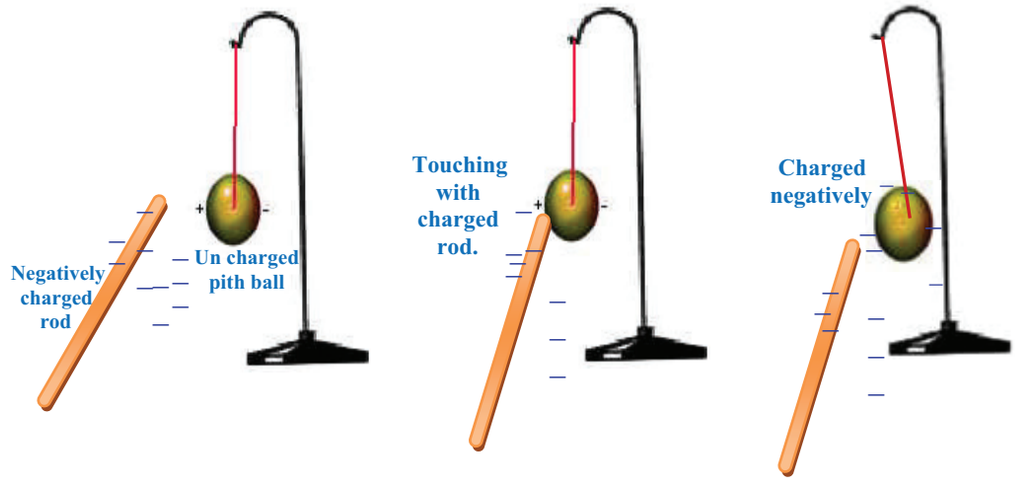


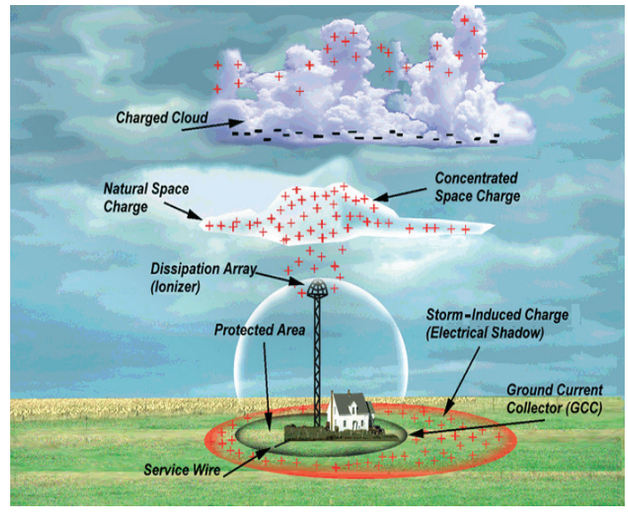
Fig 3. Charging pith ball by conduction

For example, consider a charged body that has surplus of electrons. If you bring this charged rod near the uncharged body it attracts the uncharged body. When the charged body touches the uncharged body, same negative charges transfer to the uncharged body and make it negatively charged. Then the two bodies repel each other, because they have the same charges. The process of charging a body by touching it with a charged body is called "**charging by conduction or sharing**".

Discharging a body is the process of removing electric charges from a charged body**.** A charged body can be made to lose its charges by touching it with a conductor. When a body is discharged, it becomes neutral.

Check point

1. Explain the term charging.
2. State two methods of charging a body.
3. When do we say a body is charged negatively?



3. Law of Electrostatics

Activity 3. Illustrating the law of electrostatics

***Material required*: pith balls, glass, rubber rods, woolen cloth, silk cloth**

***Procedure*:**

i. A pith ball is charged positively by conduction or touching with a charged glass rod. (Fig 4(a)
ii. Observe what happens.
iii. Repeat this activity with rubber rod rubbed with silk. Charge another pith ball by touching. (Fig 4 b)
iv. Bring two balls one touched with charged glass rod and another ball touched with charged rubber rod.
v. Bring these charged balls, closer to each other. Fig 4 (c)
vi. Observe what happens.
vii. These two balls attract each other because they carry different charges.

Conventionally negative charge is represented by (-) and positive charge is represented by (+).

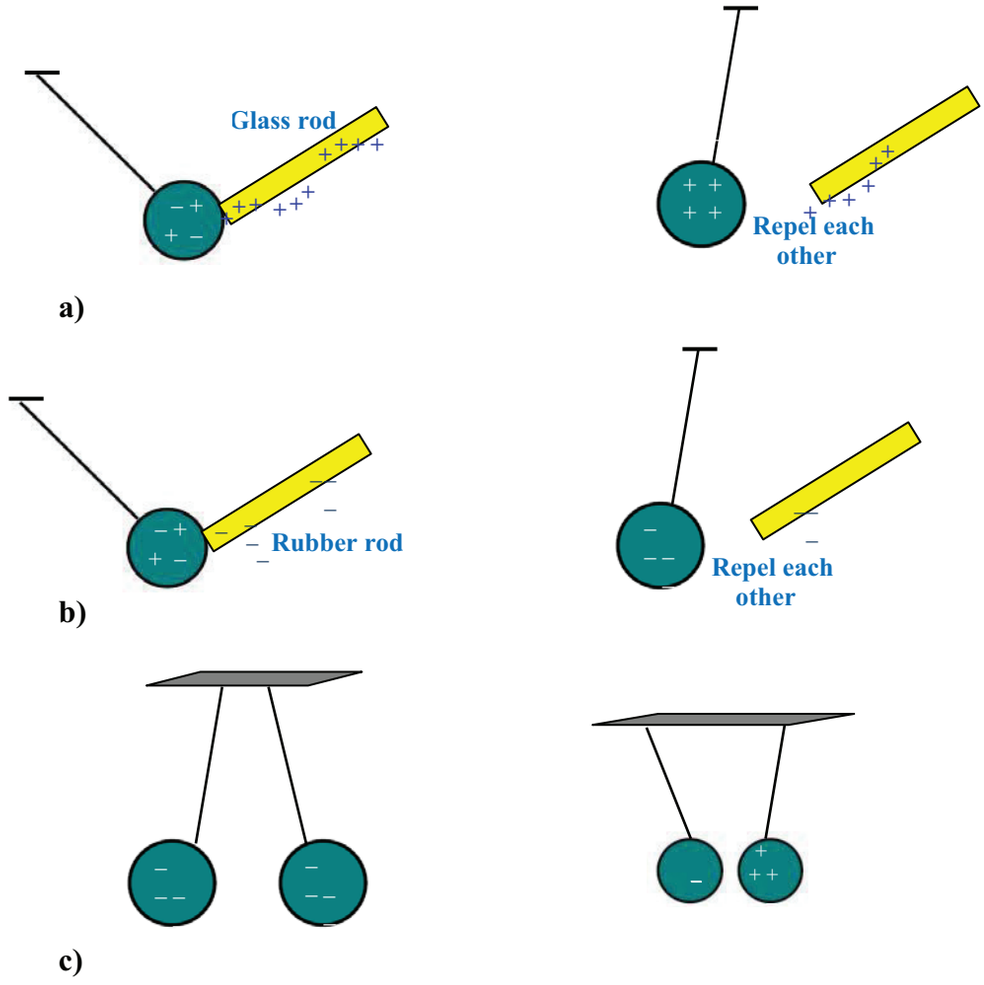


Fig 4 .Law of electrostatics

You have learnt that electric charges exert a force at a distance. It either attracts or repels other charged bodies without direct contact. Electric charges have similar properties as magnetic forces. They are non-contact forces.

All charged bodies exert forces on each other;

i) **Like charges repel each other**
ii) **Unlike charges attract each other.**
This is the law of electrostatics.

Electroscope and its use

From different activities in the previous sections, you studied how charged bodies are detected and the existence of two kinds of electric charges.

An electroscope is a simple device used to study the properties of charges. It enables us to determine both the sign of the charge and the magnitude of the charge on a body.



a) Aluminum af electroscope



b) Locally made electroscope **Fig 4 Electroscope**

Fig.4(a) shows an aluminum foil electroscope. It consists of a brass rod, on which is mounted a brass cap, at the top, and a brass plate at the bottom, with thin leaves of aluminum foil attached to the brass plate. The brass rod is mounted on a metal case, and supported by a plug of insulating material. The metal case has glass windows to allow the aluminum leaves to be seen.

4.Uses of electroscope

An electroscope is used:

i. **To illustrate that charges move through metals.**
ii. **To test whether a body is charged or not.**
iii. **To detect the sign of a charge on a body. i.e. to check whether a body is negatively or positively charged.**

Activity 4 Detecting whether a body is charged or not

Charge a glass rod by rubbing with silk clothes. Bring the charged rod near to the cap of a neutral electroscope without touching. The leaves of the neutral electroscope will diverge to show that the nearby rod is charged. The leaves will not show any deflection when uncharged rod is brought near to the cap of the neutral electroscope.

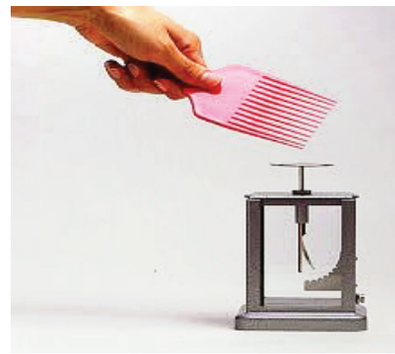
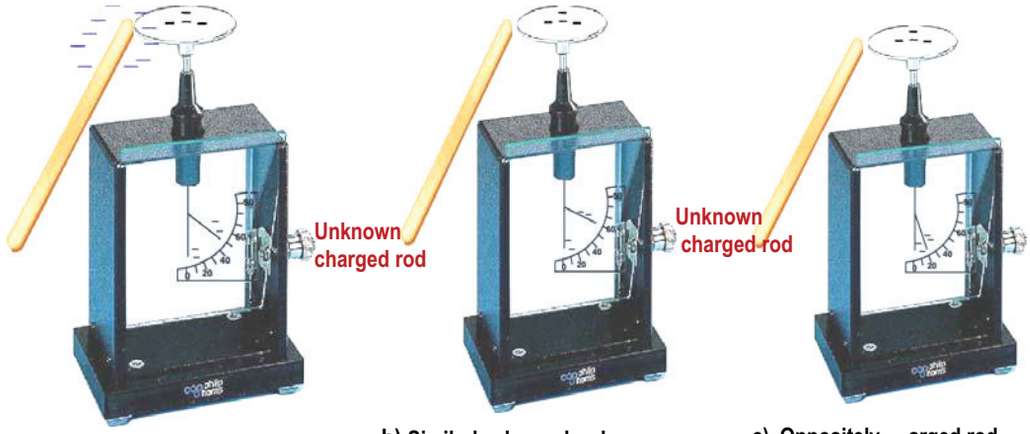


Fig 5 Deflected leaves of an electroscope

Activity 5 *Detecting the sign of charges on a charged body*

i. An electroscope must be charged first to use it to detect the sign of charges on a charged body. A neutral electroscope does not help us to detect the sign of charge on a charged body.
ii. Charge the electroscope negatively by touching its cap with a negatively charged rod. Then bring another charged rod whose sign is not known near the cap of the charged electroscope without touching it.
iii. If the divergence of the leaves increases, we can conclude that the near by charged rod have similar (--) charges as the charges of the electroscope. But if the original divergence of the leaves decreases when the unknown charged rod is brought near to the cap of the electroscope, then the sign of the unknown charged body is opposite (+) to the charge of the electroscope.



a) Charged Electroscope.

b) Similarly charged rod causes electroscope to increase in deflection.

c) Oppositely arged rod causes electroscope to decrease in deflection.

Fig 6 Detecting the sign of unknown charged body

Project work: Making your own electroscope Materials required:

- Small medicine bottle with plastic lid or cork.
- Aluminum foil obtained from the inside of a chewing gum or cigarette packet.
- Copper wire (from electric wire) or paper clip.

Procedure:

1. Bend the copper wire at one end.
2. Pierce the bottle lid with the copper wire and push the wire through it.
3. Tie the aluminum foil with silk thread or fasten it with a sellotape
4. Close the bottle with the bend of the wire at the top. This is a locally made electroscope.
5. Use this electroscope to demonstrate the uses of electroscope. For example, you can bring an uncharged body near the cap of a positively or negatively charged electroscope. What do you think will happen to the leaves? Will the leaves decrease or increase its divergence?

5. Danger of Static electricity

a. Lightning

A tall building is protected by a lightning conductor consisting of a thick copper strip fixed on the outside of the building connecting metal spikes at the top to a metal plate in the ground (Figure 7). Thunderclouds carry charges; a negatively charged cloud passing overhead repels electrons from the spikes to the Earth. The points of the spikes are left with a large positive charge (charge concentrates on sharp points) which removes electrons from nearby air molecules, so charging them positively and causing them to be repelled from the spike.

This effect, called action at points, results in an 'electric wind' of positive air molecules streaming upwards which can neutralize electrons discharging from the thundercloud in a lightning flash. If a flash occurs, it is now less violent and the conductor gives it an easy path to ground.

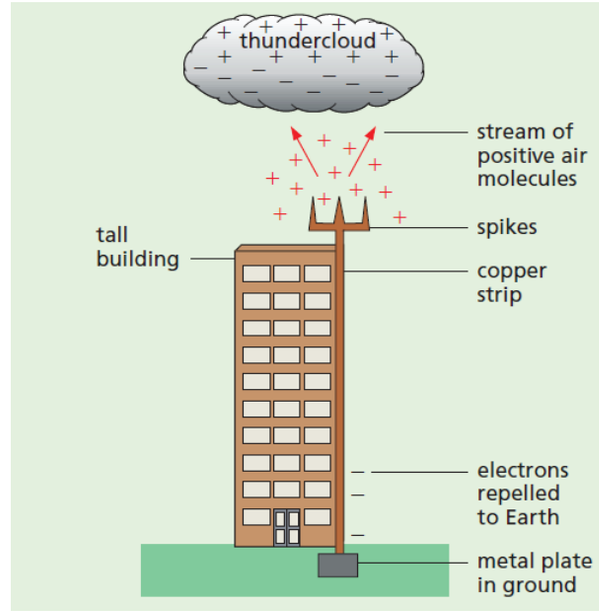


Fig.7 Lightening rod



Fig.8 Safe aircraft refueling

b. Refueling

Sparks from static electricity can be dangerous when flammable vapor is present. For this reason, the tanks in an oil tanker may be cleaned in an atmosphere of nitrogen –otherwise oxygen in the air could promote a fire. An aircraft in flight may become charged by 'rubbing' the air. Its tyres are made of conducting rubber which lets the charge pass harmlessly to ground on landing, otherwise an explosion could be 'sparked off' when the aircraft refuels. What precautions are taken at petrol pumps when a car is refueled?

Questions

i. Multiple Choices

9. Which of the following is an example of static electricity?
   a. A battery
   b. A person being shocked when touching a doorknob
   c. A comb picking up paper scraps
   d. An electricity generating station

10. A proton has a positive charge. What is the charge of a particle that is attracted to a proton?
   a. Negative
   b. Neutral
   c. Positive
   d. Repulsive

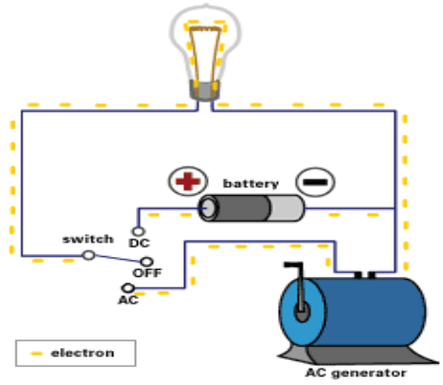
11. When a light is on, it is part of a _____ circuit.
   a. Open
   b. Closed
   c. Big
   d. Little

12. A light bulb converts electrical energy into which of the following?
   a. Light only
   b. Heat only
   c. Heat and light
   d. Light and motion

LESSON 2: ELECTRIC CURRENT

At the end of this chapter, you should be able to:

- state that there are positive and negative charges and that charges is measured in Coulomb.
- state that current is a rate of flow of charge and that it is measured in amperes.
- distinguish between conventional current and electron flow
- recall and apply the relationship, charge= current x time.



1.Electric Charges

Atoms are made up of protons, neutrons and electrons. The protons (positive) and electrons (negative) in matter are the carriers of electric charge. The attraction of unlike charges and repulsion of the like charges cause electrical phenomena. Charge is measured in coulomb, C.

2. Conventional Current and Electron Flow

Electricity refers to the flow of a stream of charged particles. Let's take a closer look at the inside of a piece of metal wire.

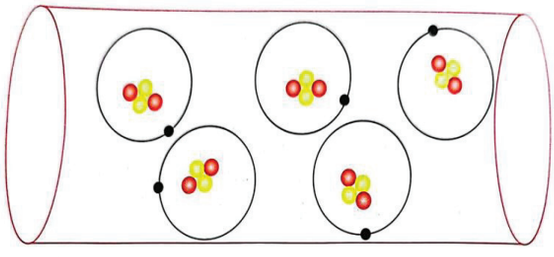


Fig 1a. Free electron moving around the nucleus

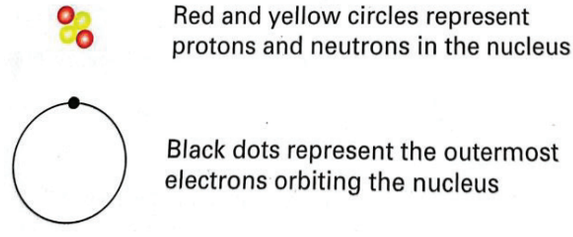


Fig 1b. Representation

A metal is made up of atoms. Each atom consists of a nucleus and electrons. The outermost electrons orbit the nucleus, sometimes wandering off to nearby atoms randomly. Electrons, which carry a negative charge, flow from negative terminal to positive terminal.

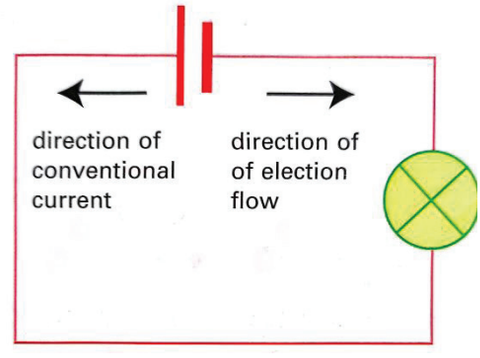


Fig 2. Direction of electrons and current

Conventional current is the flow of the charges from positive terminal to negative terminal. Positively charged particles are involved in the conventional current. However, electrons move in the opposite direction as shown in Figure. Conventionally, the direction of the electric current is taken to be the flow of positive charge.

3.Electric Current

Electric current, I is defined as the rate of flow of charges. The SI unit for current is ampere (A). The amount of charge passing any point in a circuit depends on the current, I and the time taken for the charge to pass the point. The amount of charge Q is the product of current and time, that is

$$Q = I \times t$$ where Q *is the amount of charge in Coulomb (C),*
I *is the current in ampere (A),*
t *is the time taken in second (s).*

One coulomb is the charge passing any point in a circuit when a steady current of 1A flows for 1 second. That is, 1 C = 1 A s.

Worked example:

A current of 4A flows through a light bulb. Find the amount of charge flowing through the Bulb in 60s.

Solution:

Find the amount of charge

Using: Q = I t Where I = 4A, t = 60s

Therefore, Q = 4x60 = 24 C

4. Circuit Diagrams

Current must have a complete path (a circuit) of conductors if it is to flow. Wires of copper are used to connect batteries, lamps, etc. in a circuit since copper is a good electrical conductor. If the wires are covered with insulation, such as plastic, the ends are bared for connecting up.

The signs or symbols used for various parts of an electric circuit are shown in Figure 36.3.

Before the electron was discovered scientists agreed to think of current as positive charges moving round a circuit in the direction from positive to negative of a battery. This agreement still stands. Arrows on circuit diagrams show the direction of what we call the conventional current, i.e. the direction in which positive charges would flow. Electrons flow in the opposite direction to the conventional current.

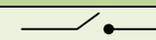
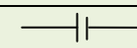
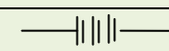
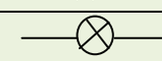
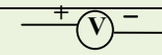
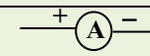
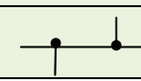
Items	Symbols
Switch	
Cell	
Battery	
Lamp	
Resistor	
Voltmeter	
Ammeter	
Crossing wires	
Joined wires	

Fig3. Symbols of elements of an electric circuit

Practical work (Measuring current)

(a) Connect the circuit of Figure 4 (on a circuit board if possible) ensuring that the + of the cell (the metal stud) goes to the + of the ammeter (marked red). Note the current.

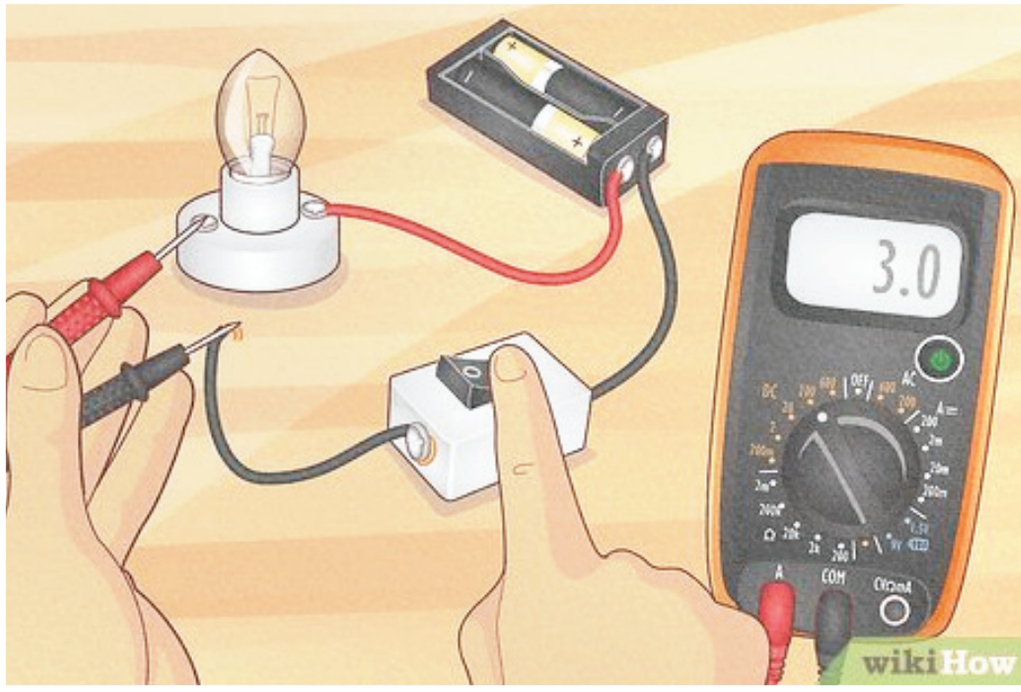
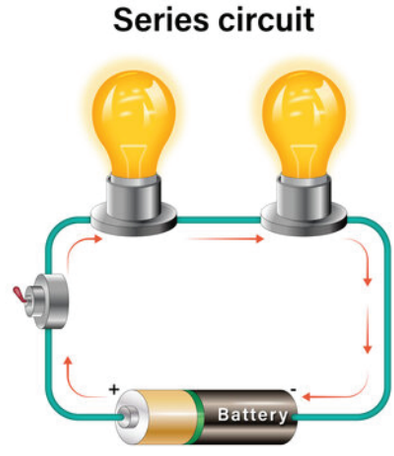


Fig 4. Symbols of elements of an electric circuit

5. Series and parallel circuits



a. Series

In a series circuit, the current that goes through one lamp goes through the other lamp as well. So, the current that goes through lamp X is the current that goes through lamp Y. When one of the lamps blows, the other lamp will not light up even though the circuit is closed.

The current is the same at all points in a series circuit.

An Ammeter must be connected such that the positive terminal of the ammeter is always connected To the positive terminal of the cell.

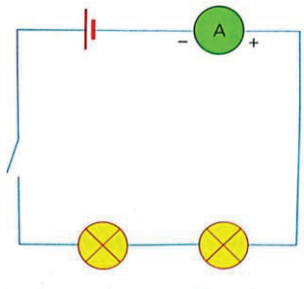
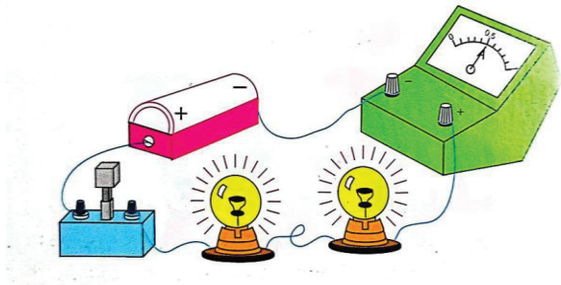


Fig. 12.9a An ammeter connected in series with the circuit

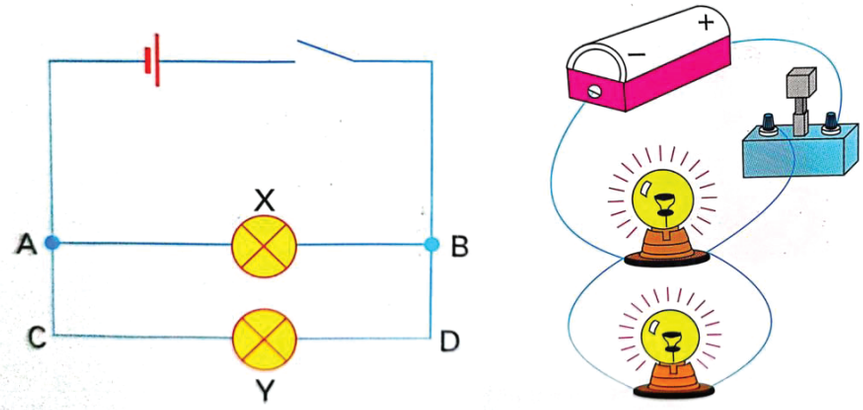


12.9b The positive terminal of ammeter connected to the positive terminal of cell

b. Parallel

In a parallel circuit, as in Figure below, the lamps are side by side and there are alternative paths for the current. The current splits: some goes through one lamp and the rest through the other. The current from the source is larger than the current in each branch.

The sum of the currents in the branches of a parallel circuit equals the current entering or leaving the parallel section.





End of Lesson

PROBLEMS

1. (a) Write down the symbol and SI units for charge and current in the table below.

Quantity		SI unit	
Name	Symbol	Name	Symbol
Charge			
Current			

(b) State a relationship between charge, current and time,

(c) Complete the table below.

Charge/C	Current/A	Time/s
	2.0	3.5
33	45	

2. Find the unknown quantity (CONVERT FIRST to seconds)

a) I = Q = 140 C t = 4 min = _____ s	b) I = 0.3 A Q = t = 1.5 hours = _____ s	c) I = 0.9 A Q = t = 3 min = _____ s

3. Find the unknown quantity (CONVERT FIRST to SECONDS)

a) I = Q = 140 C t = 4 min = _____ s	b) I = 0.3 A Q = t = 1.5 hours = _____ s	c) I = 0.9 A Q = t = 3 min = _____ s
d) I = Q = 200 C t = 5 min = _____ s	e) I = 1.5 A Q = t = 2 hours = _____ s	f) I = 0.4 A Q = t = 7 min = _____ s

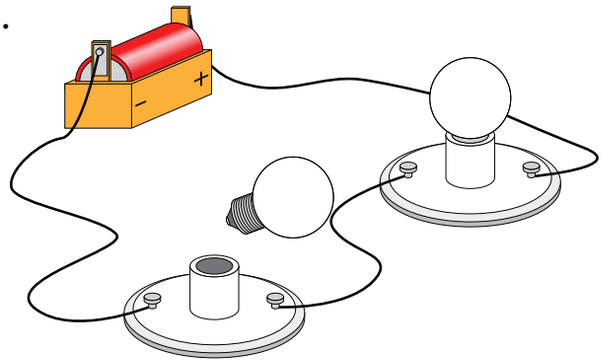
4. How much current must there be in a circuit if 1000 Coulombs flow past a point in the circuit in 4 minutes?
5. A circuit is switched on for half a minute and 90 Coulombs of charge flowed. What was the current flowing through the circuit?
6. If there is a current of 10 mA in a circuit for 0.5 s, what quantity of electric charge flows in through the circuit?
7. How much time is required for 0.3 Coulombs of charge to flow past a point if the rate of flow (current) is 2 mA?



Electrical Circuits

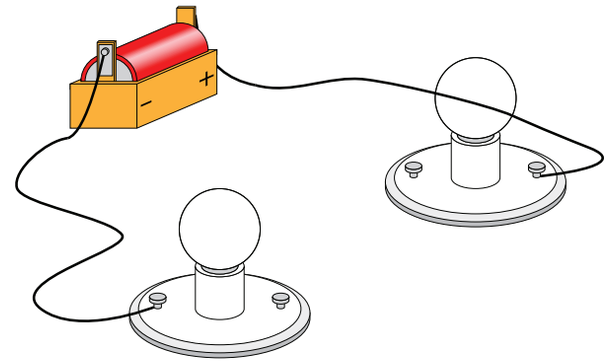
Tell whether the light bulb or bulbs **will light** or **will not light** based on the circuit.

1.



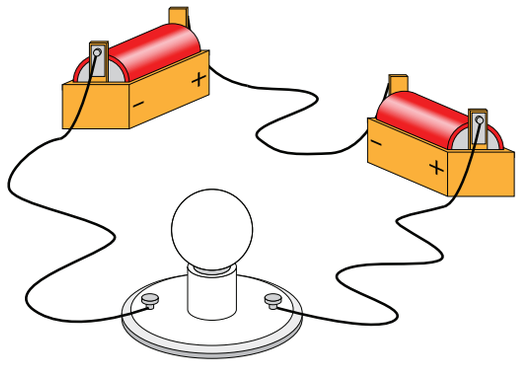
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2.



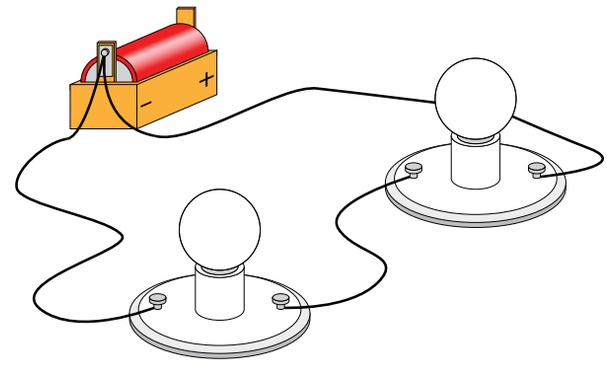
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3.



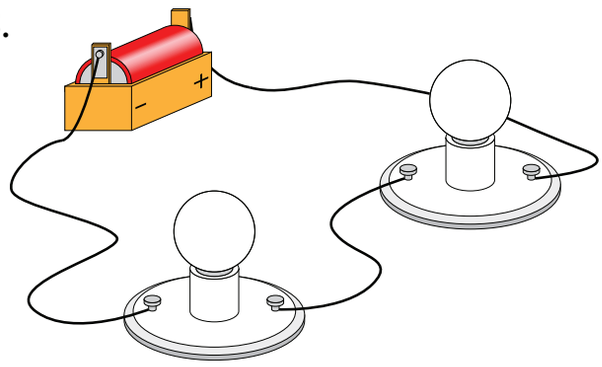
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4.



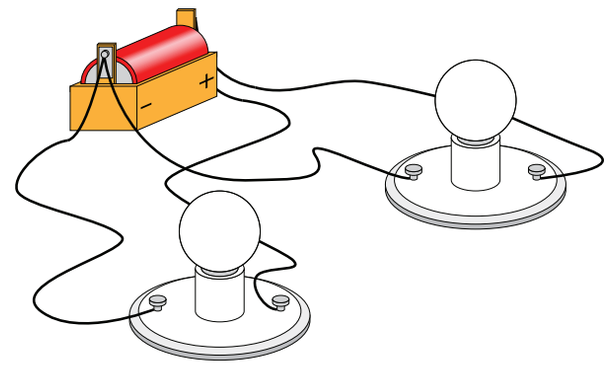
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5.



_____

6.

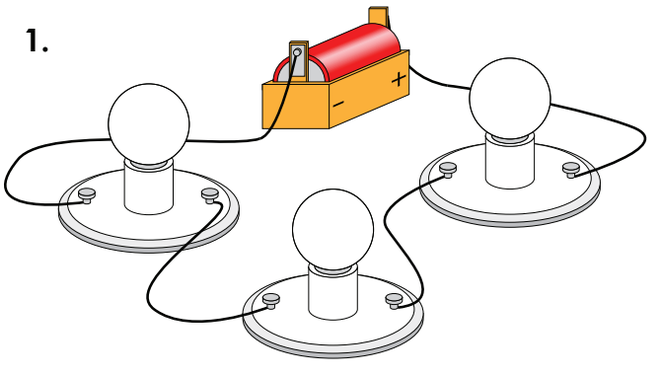


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Series & Parallel Circuits

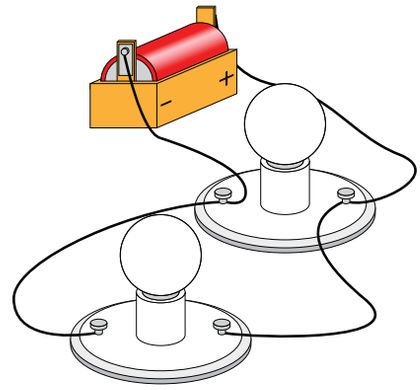
Tell whether each picture shows a **series circuit** or **parallel circuit**.

1.



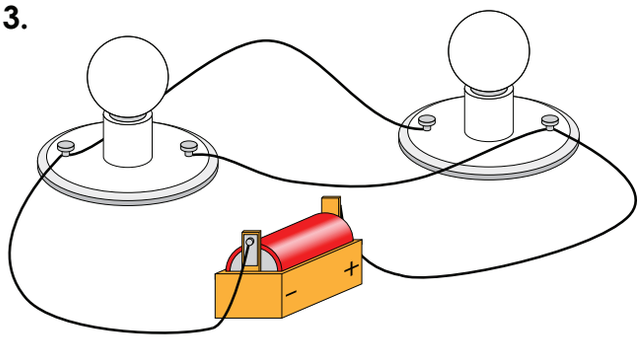
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2.



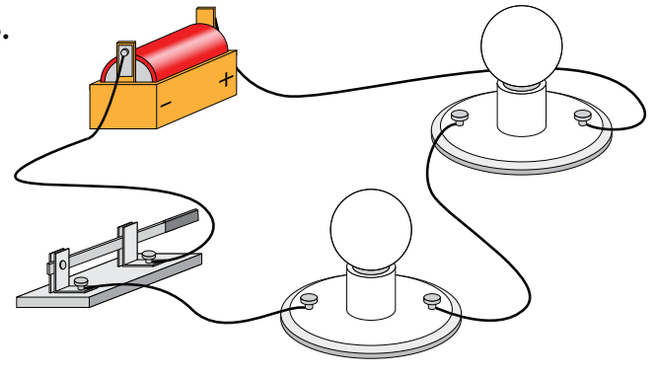
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3.



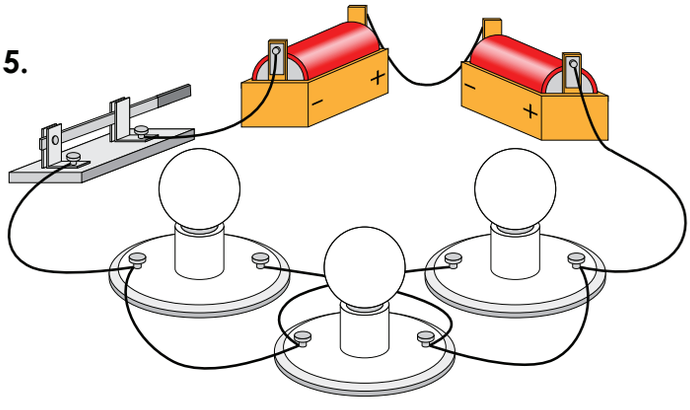
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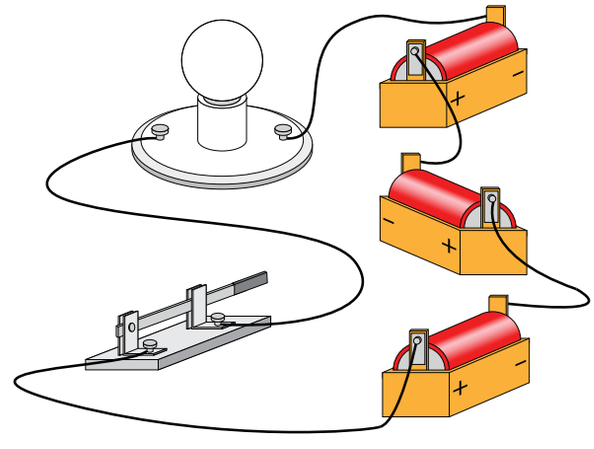
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5.



type: _____

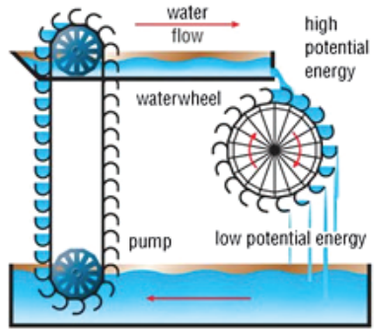
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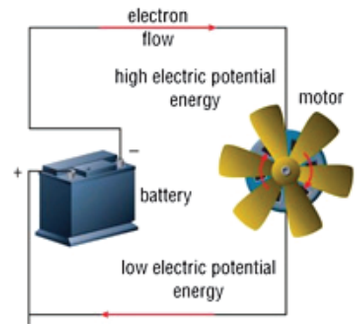


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LESSON 3

ELECTRIC POTENTIAL DIFFERENCE





At the end of this chapter, you should be able to:

- recognize that a potential difference, when across a wire or a component, is what causes a current,
- recall that potential difference is measured in volts,
- recall that the potential difference is the work that would have to be done per coulomb in order to separate the positive and negative charges.

A battery transforms chemical energy to electrical energy. Because of the chemical action going on inside it, it builds up a surplus of electrons at one of its terminals (the negative) and creates a shortage at the other (the positive). It is then able to maintain a flow of electrons, i.e. an electric current, in any circuit connected across its terminals for as long as the chemical action lasts.

The battery is said to have a potential difference (p.d. for short) at its terminals. Potential difference is measured in volts (V) and the term voltage is sometimes used instead of p.d. The p.d. of a car battery is 12 V and the domestic mains supply is 220V.

1. Energy transfer and p.d.

In an electric circuit electrical energy is supplied from a source such as a battery and is transferred to other forms of energy by devices in the circuit. A lamp produces heat and light.

When each one of the circuits of Figure 1 is connected up, it will be found from the ammeter readings that the current is about the same (0.4 A) in each lamp. However, the mains lamp with a potential difference of 220 V applied to it gives much more light and heat than the car lamp with 12 V across it. In terms of energy, the mains lamp transfers a great deal more electrical energy in a second than the car lamp.

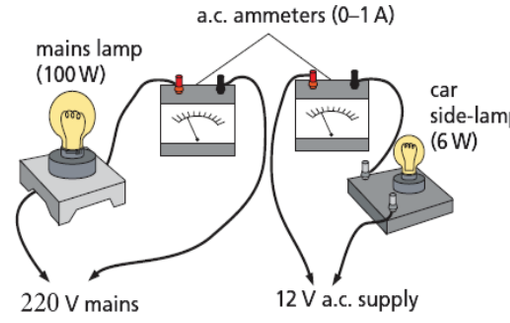


Figure 1 Investigating the effect of p.d. (potential difference) on energy transfer

Evidently the p.d. across a device affects the rate at which it transfers electrical energy. This gives us a way of defining the unit of potential difference: the volt.

2.The volt

The demonstrations of Figure1 show that the greater the voltage at the terminals of a supply, the larger is the 'bundle' of electrical energy given to each coulomb and the greater is the rate at which light and heat are produced in a lamp.

The p.d. between two points in a circuit is 1 volt if 1 joule of electrical energy is transferred to other forms of energy when 1 coulomb passes from one point to the other.

That is, 1 volt = 1 joule per coulomb (1 V = 1 J/C). If 2 J are given up by each coulomb, the p.d. is 2 V. If 6 J are transferred when 2 C pass, the p.d. is 6 J/2 C = 3 V.

In general, if E (joules) is the energy transferred (i.e. the work done) when charge Q (coulombs) passes between two points, the p.d. V (volts) between the points is given by

$$\mathbf{V = \frac{E}{Q}} \quad \text{or} \quad \mathbf{E = V \times Q}$$

If Q is in the form of a steady current I (amperes) flowing for time t (seconds) then

$$\mathbf{Q = I \times t} \qquad \boldsymbol{E = I \times V \times t}$$

3. Cells and battery

The common sources of potential difference are electric **cells** and **generators**. **Battery** is a combination of two or more electric cells. The battery holds electric charge and has a voltage which can be thought of as like water pressure.

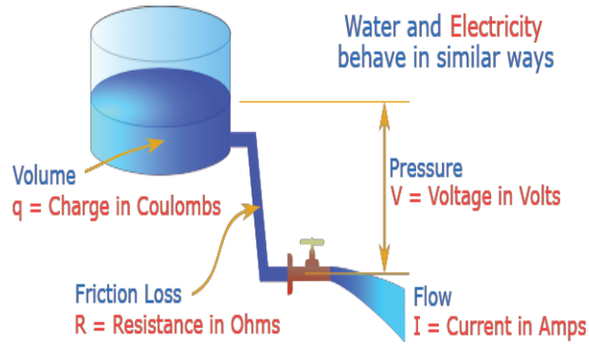
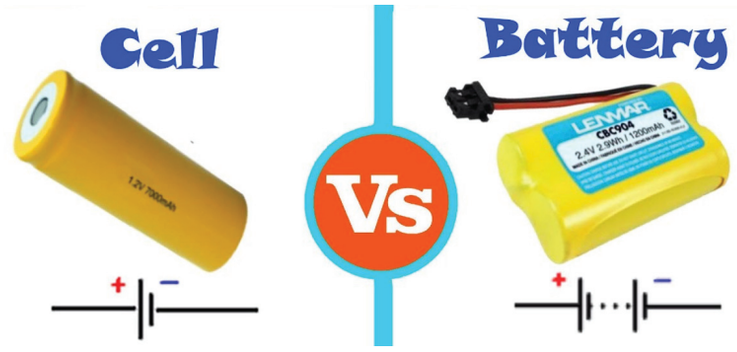
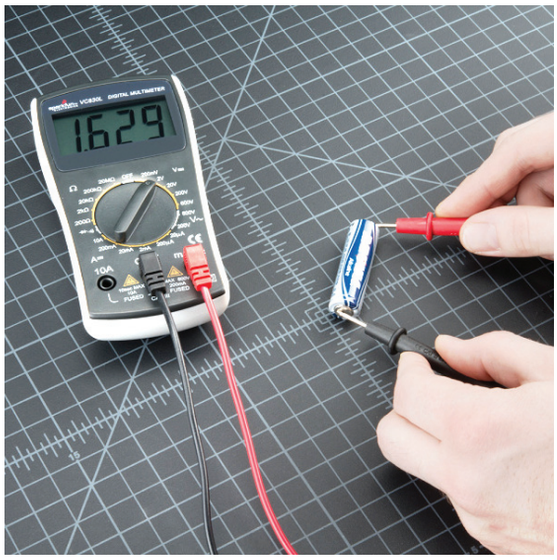


Fig. 3. This is the Electricity Water Analogy

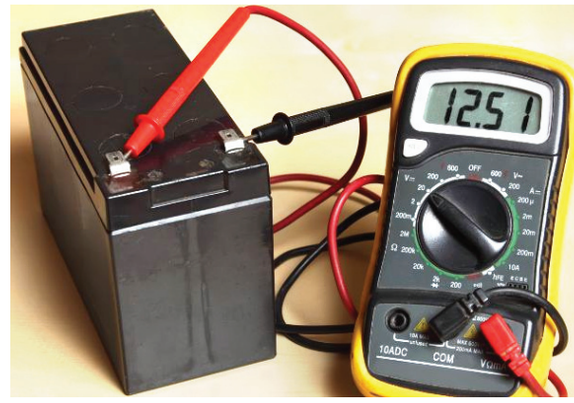


4.Measuring cell and battery voltage with a voltmeter

The potential difference across any two terminals is measured by an instrument called **voltmeter**. In using a voltmeter:



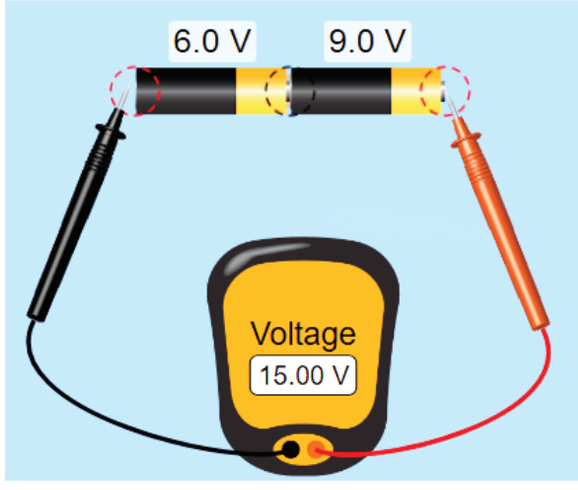
- Connect it in parallel to a battery to measure the voltage across the battery.
- Red lead is connected to the positive of the battery as in Fig.
- Black lead is connected to the negative of the battery as in Fig.

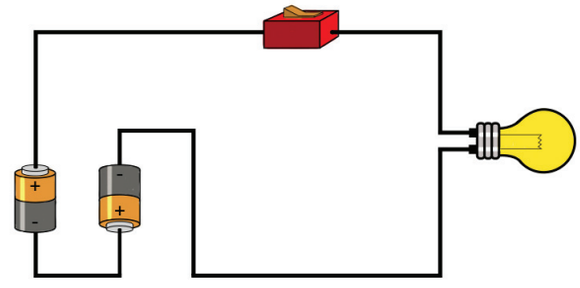


5.Cells in series

Connecting batteries in series increases the voltage of a battery pack. To connect batteries in a series, a jumper wire connects a battery's negative terminal to another battery's positive terminal. This leaves you with a positive terminal at the beginning of the battery pack and a negative terminal at the end of the battery pack for your application.

For example, the image below shows four 1.5-volt cells wired in series, producing a 6-volt battery pack. Remember, only the voltage goes up in series.



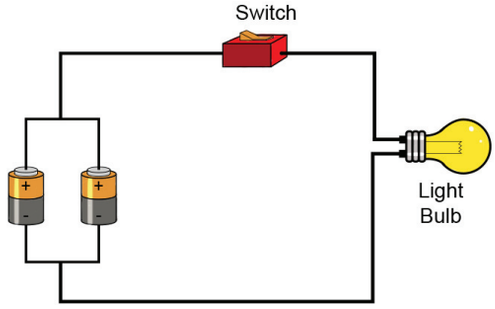


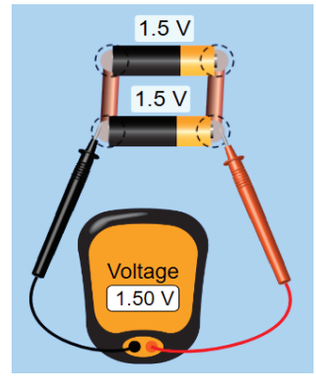
The total voltage the batteries is the sum of voltage of all cells. $V=V_1+V_2+V_3+...$

6.Cells in parallel

The other type of connection is parallel. Parallel connections will increase your capacity rating, but the voltage will stay the same. In the "Parallel" diagram, we're back to 12 volts..

To join batteries in parallel, use a jumper wire to connect positive terminals together, and another jumper wire to connect negative terminals together. This establishes negatives to negatives and positives to positives. You CAN connect your load to ONE of the batteries, which will drain both equally. However, the preferred method for keeping the batteries equalized is connecting to the positive at one end of the battery pack and the negative at the other end.



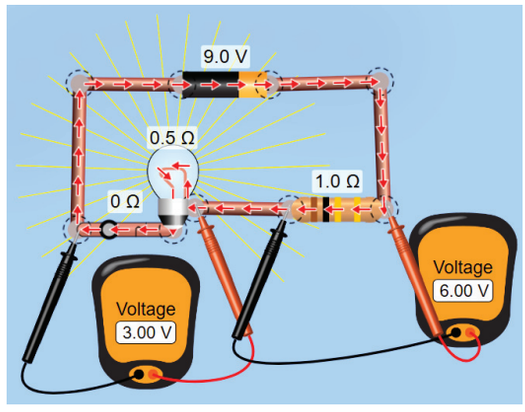


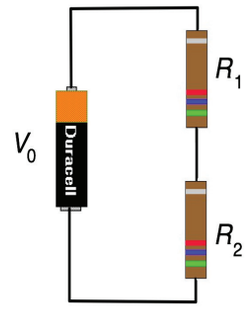
The total voltage across the batteries is equal to the voltage of each cell:

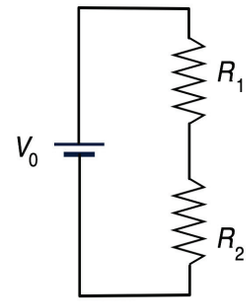
$$V=V_1=V_2=V_3$$

7. Voltages round a circuit

A.Voltage across Series Circuit







In the experiment you should have found in the circuit that:

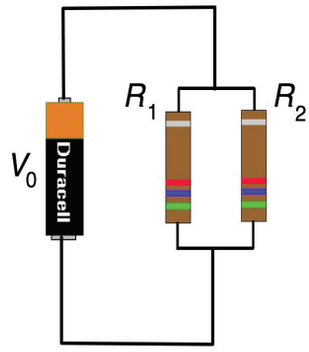
$$V = V1 + V2$$

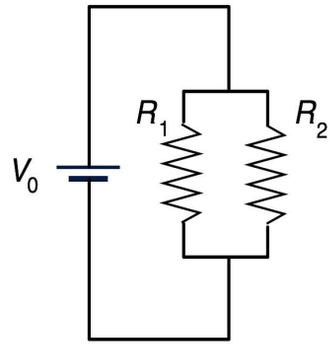
The voltage at the terminals of a battery equals the sum of the voltages across the devices in the external circuit from one battery terminal to the other.

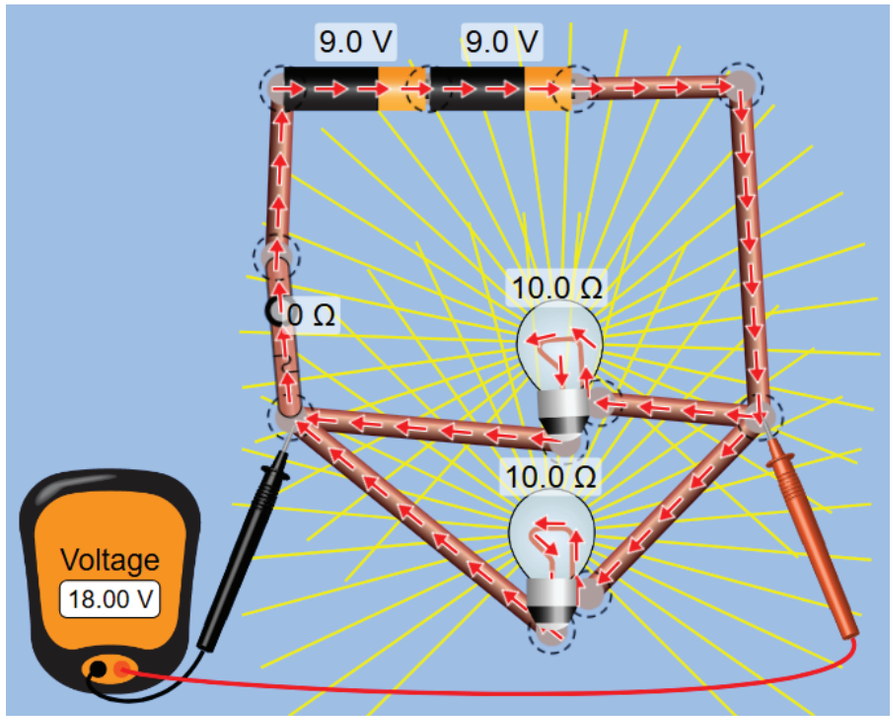
B. Voltage across a Parallel Circuit

In the experiment you should have found in the circuit that:

The voltages across devices in parallel in a circuit are equal. $V=V_1=V_2$









PROBLEMS

1. The p.d. across the lamp in Figure 6 is 12 V. How many joules of electrical energy are changed into light and heat when a. a charge of 1 C passes through it, b. a charge of 5 C passes through it, c. a current of 2 A flows in it for 10 s?

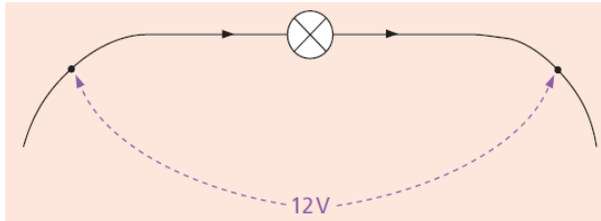


Figure 6

2. Three 2-V cells are connected in series and used as the supply for a circuit. a What is the p.d. at the terminals of the supply? b How many joules of electrical energy does 1 C gain on passing through a. One cell b. all three cells?

3. Each of the cells shown in Figure 7 has a p.d. of 1.5 V. Which of the arrangements would produce a battery with a p.d. of 6 V?

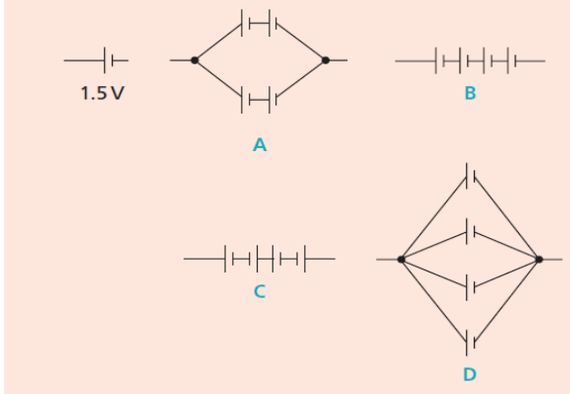


Figure 7

4. The lamps and the cells in all the circuits of Figure 8 are the same. If the lamp in a has its full, normal brightness, what can you say about the brightness of the lamps in b, c, d, e and f?

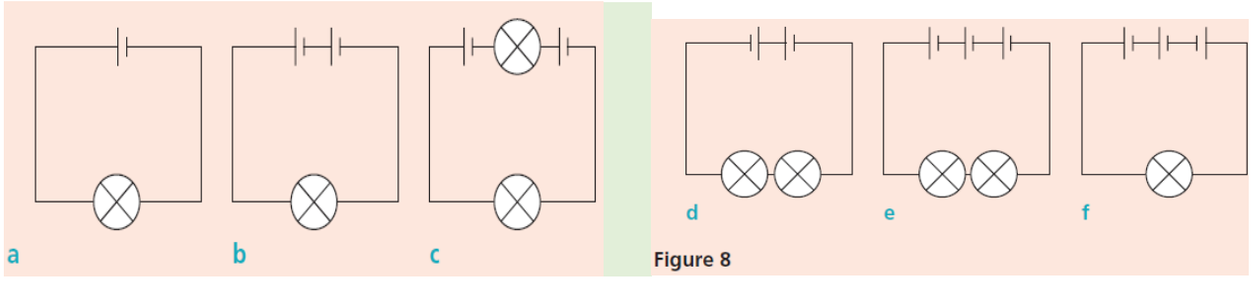


Figure 8

5. Three voltmeters V, V1 and V2 are connected as in Figure 9.
   a. If V reads 18 V and V1 reads 12 V, what does V2 read?
   b. If the ammeter A reads 0.5 A, how much electrical energy is changed to heat and light in lamp L1 in one minute?
   c. Copy Figure 9 and mark with a + the positive terminals of the ammeter and voltmeters for correct connection.

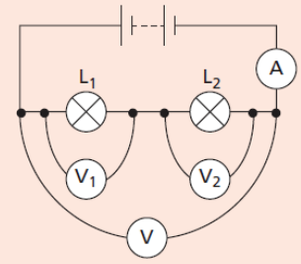


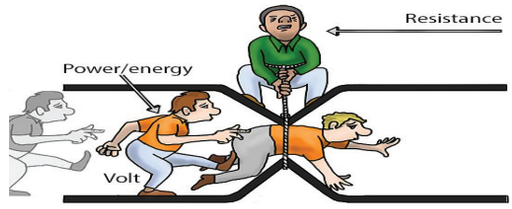
Figure 9

6. Calculate the energy transferred in a component when the charge passing through it is 30 C and the potential difference is 20 V.
7. Calculate the energy transferred when the charge flow is 30 C and the potential difference is 4 V.
8. Calculate the energy transfer for a charge flow of 20 C when the potential difference is 6.0 V.
9. Calculate the voltage supplied to a 5,000 J appliance that transfers 20 C of electrical charge.
10. Calculate the charge transferred by a 5000 J electrical appliance when the voltage supplied to it is 230 V.
11. Calculate the voltage supplied to a 0.05 MJ appliance that transfers 3 C of electrical charge.



Lesson 4

ELECTRIC RESISTANCE



At the end of this chapter, you should be able to:

- describe the properties of resistance
- state the Ohm's law
- determine the resistance using Ohm's law
- distinguish between resistance and resistor
- describe types of resistors
- draw I-V diagram
- apply the relationship of the proportionality of resistance to the length and cross-sectional area of a wire to solve problems.

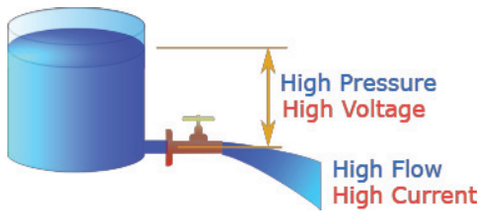
Electrons move more easily through some conductors than others when a p.d. is applied. The opposition of a conductor to current is called its **resistance**. A good conductor has a low resistance and a poor conductor has a high resistance.

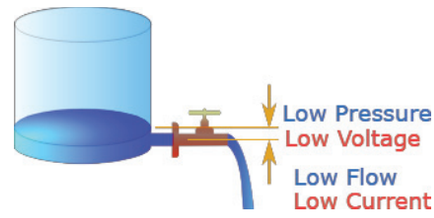
1 . The Ohm

If the current in a conductor is I when the voltage across it is V, as shown in Figure 1a, its resistance R is defined by

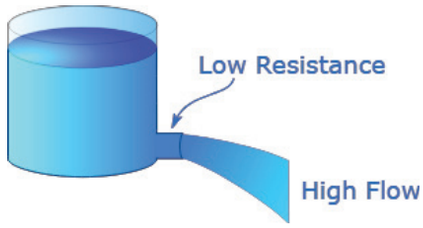
$$R = \frac{V}{I}$$

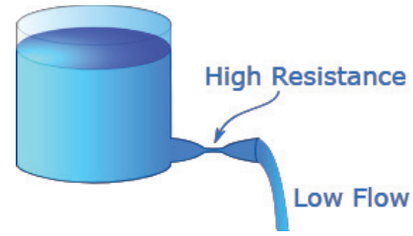
where R is the resistance of the conductor in (Ω)
V is the P.d across the conductor in (V)
I is the current flowing through the conductor in (A).





The ohm is the resistance of a conductor in which the current is 1 ampere when a voltage of 1 volt is applied across it.





2. Resistors

Conductors intended to have resistance are called resistors (Figure 2a) and are made either from wires of special alloys or from carbon. Those used in radio and television sets have values from a few ohms up to millions of ohms (Figure 2b).



Figure 2a Circuit symbol for a resistor



Figure 2b Resistor



Figure 2c Variable resistor (potentiometer)

Variable resistors are used in electronics (and are then called potentiometers) as volume and other controls (Figure 2c). Variable resistors that take larger currents, like the one shown in Figure 3, are useful in laboratory experiments. These consist of a coil of constantan wire (an alloy of 60% copper, 40% nickel) wound on a tube with a sliding contact on a metal bar above the tube.

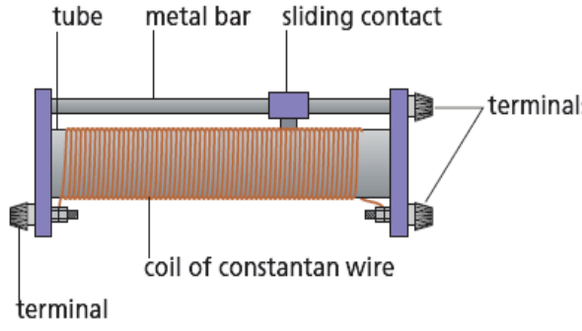


Figure 3 Large variable resistor

There are two ways of using such a variable resistor. It may be used as a rheostat for changing the current in a circuit; only one end connection and the sliding contact are then required. In Figure 4a moving the sliding contact to the left reduces the resistance and increases the current. This variable resistor can also act as a potential divider for changing the p.d. applied to a device; all three connections are then used. In Figure 4b any fraction from the total p.d. of the battery to zero can be 'tapped off' by moving the sliding contact down. Figure 5 shows the circuit diagram symbol for a variable resistor being used in rheostat mode.

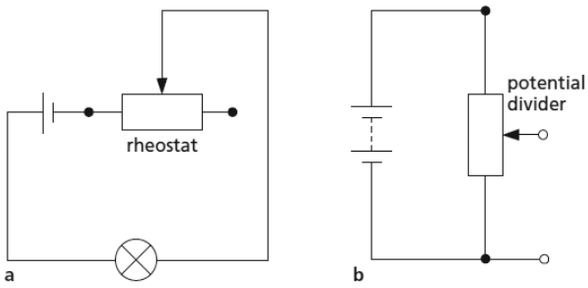


Figure 4 A variable resistor can be used as a rheostat or as a potential divider.

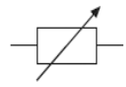


Figure 5 Circuit symbol for a variable resistor used as a rheostat

Practical work

Measuring resistance

The resistance R of a conductor can be found by measuring the current I in it when a p.d. V is applied across it and then using R = V/I. This is called the ammeter–voltmeter method.

Set up the circuit of Figure 6 in which the unknown resistance R is 1 meter of **Standard Wire Gauge** or **SWG**) 34 constantan wire. Altering the rheostat changes both the p.d. V and the current I. Record in a table, with three columns, five values of I (e.g. 0.10, 0.15, 0.20, 0.25 and 0.3 A) and the corresponding values of V. Work out R for each pair of readings. Repeat the experiment, but instead of the wire use a lamp (e.g. 2.5 V, 0.3 A).

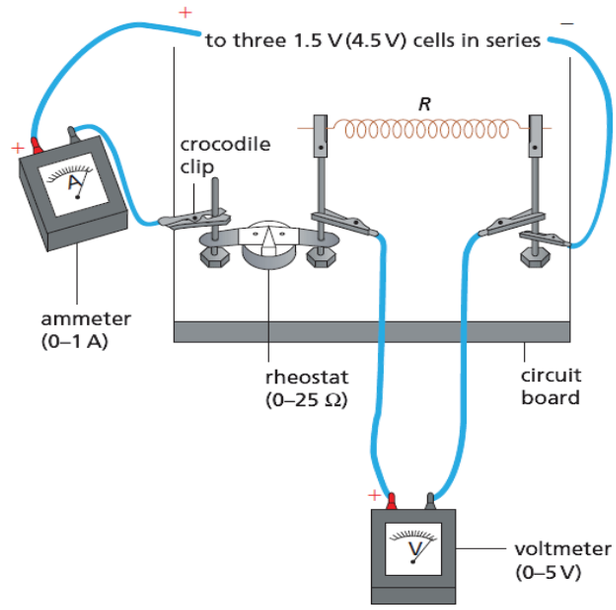
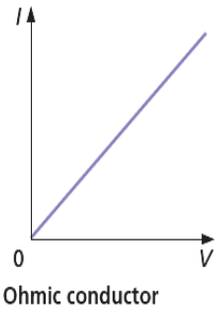


Figure 6

3. The I–V graphs: Ohm's law

The results of the experiment allow graphs of I against V to be plotted for different conductors.

Metallic conductors



a Ohmic conductor

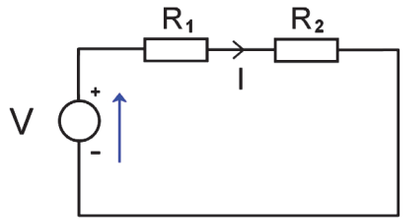
Metals and some alloys give I–V graphs that are a straight line through the origin, as in Figure 7a, provided that their temperature is constant. I is directly proportional to V, i.e. I ∝ V. Doubling V doubles I, etc. Such conductors obey Ohm's law, stated as follows.

The current in a metallic conductor is directly proportional to the p.d. across its ends if the temperature and other conditions are constant.

They are called Ohmic or linear conductors and since I ∝ V, it follows that V/I = a constant (obtained from the slope of the I–V graph). The resistance of an Ohmic conductor therefore does not change when the p.d. does.

4. Two Resistors in series

The resistors in Figure are in series. The same current I flows through each and the total voltage V across all three is the sum of the separate voltages across them, i.e. $V = V_1 + V_2$



But $V_1 = IR_1$,$V_2 = IR_2$

Also, if R is the combined resistance, V = IR, and so $IR = IR_1 + IR_2$

Dividing both sides by I,

$$R = R_1 + R_2$$

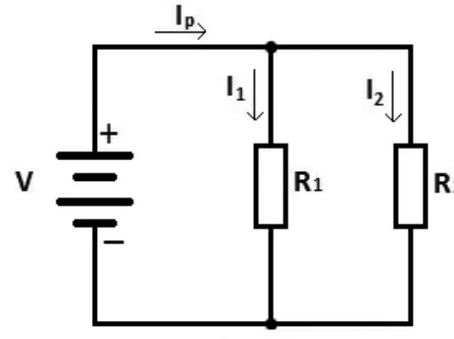
Example: What is the total resistance here:



R_{tot} = 9 Ω + 9 Ω + 9 Ω = 27 Ω

5. Two Resistors in parallel

The resistors in Figure 10 are in parallel. The voltage V between the ends of each is the same and the total current I equals the sum of the currents in the separate branches, i.e. $I = I_1 + I_2$



$$I_1 = \frac{V}{R_1}, \qquad I_2 = \frac{V}{R_2}$$

Also, if R is the combined resistance, $I = \frac{V}{R}$,

Then $\frac{V}{R} = \frac{V}{R_1} + \frac{V}{R_2}$

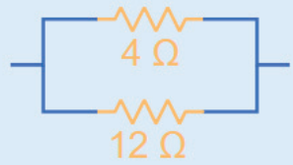
Dividing both sides by V, $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$

For the simpler case of two resistors in parallel: $\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{R_2}{R_1 \times R_2} + \frac{R_1}{R_1 \times R_2} = \frac{R_1 + R_2}{R_1 \times R_2}$

Inverting both sides, $R = \frac{R_1 \times R_2}{R_1 + R_2}$

The combined resistance of two resistors in parallel is less than the value of either resistor alone. Check this is true in the following Worked example. Lamps are connected in parallel rather than in series in a lighting circuit. Can you suggest why?

Example: What is the total resistance here:



$$\frac{1}{R_{tot}} = \frac{1}{4} + \frac{1}{12}$$

$$\frac{1}{R_{tot}} = \frac{3}{12} + \frac{1}{12}$$

$$\frac{1}{R_{tot}} = \frac{4}{12} = \frac{1}{3}$$

So $R_{tot} = 3\ \Omega$

Or we can do it in one go (using a calculator):

$$R_{tot} = 1/(\frac{1}{4} + \frac{1}{12}) = 3\ \Omega$$



End of Lesson

Problems

I. Multiple Choice Questions

1. Which one opposes the current?
   a. Resistor b. voltage
   c. Inductor d. Coil

2. Which of the following is the symbol for current?
   a. A b. V
   c. Ω d. I

3. A volt is a/an:
   a. Potential difference between two points b. Unit of charge on an electron
   c. Rate of flow of charge d. Opposition to current flow

4. How many forms of Ohm's Law are there?
   a. One b. Two
   c. Three d. Four

5. When the voltage across a resister is doubled, the current will:
   a. Triple b. Halve
   c. Double d. Not change

6. When the resistance in a circuit is doubled, the current will:
   a. Triple b. Halve
   c. Double d. Not change

7. What is the definition of current?
   a. The movement of electrons b. Force created by the buildup of charge
   c. Movement of wire d. Movement of molecule

8. What is resistance?
   a. The interaction of two magnetic fields
   b. Conversion of mechanical energy into electrical energy
   c. An imbalance between electrons and protons
   d. The opposition to current flow

9. What is the formula for Ohm's Law?
   a. $P = V I$ b. $I = V / R$
   c. $F = M A$ d. $I = V^2 R$

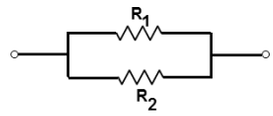
10. The unit of electrical charge is the:
    a. Coulomb b. Joule
    c. Ampere d. Ohm

11. The unit of resistance is the:
    a. Coulomb b. Joule
    c. Ampere d. Ohm

12. An ammeter connected in series with three resistors reads an electric current of 2 A. What is the electric current flowing through resistor R_3?
    a.1 A b. 2 A
    c. 3 A d. 4 A



13. Two resistor $_1$ = 3 Ω and R_2 = 6 Ω are connected in parallel. What is the net resistance in the circuit?



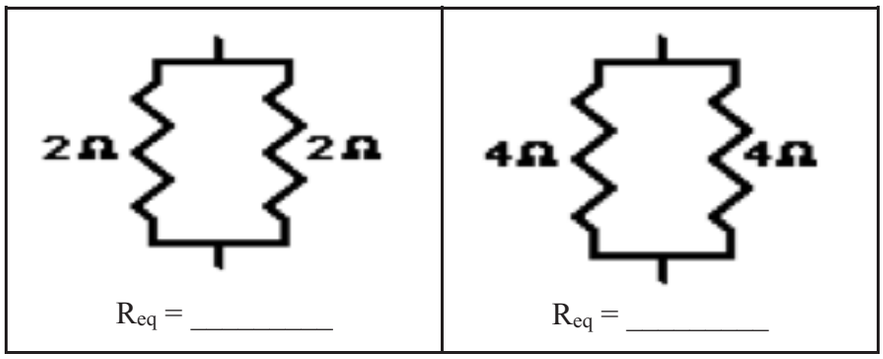
    a. 1 Ω b. 3 Ω
    c. 6 Ω d. 2 Ω

14. Convert

200mA= ……A	0.5kA=…….A	100A=……...kA
500mΩ= ……..Ω	100kΩ= …… Ω	1000Ω=…….kΩ
3000mV=……V	380kV=……V	240V =……..kV

II. Exercise

1. State the Ohm's Law
2. A circuit has a 12 V power supply and a 1 kΩ resistor. What is the current?
3. A circuit has a 42 kΩ resistor and 8 mA of current. What is the voltage?
4. A circuit has 520 mA of current and 240 volts. What is the resistance?
5. Write the main equation of Ohm's law
6. What is the resistance of a lamp when a voltage of 12 V across it causes a current of 4 A?
7. Calculate the p.d. across a 10 Ω resistor carrying a current of 2 A.
8. Solve the following for their equivalent resistance (equivalent resistance).



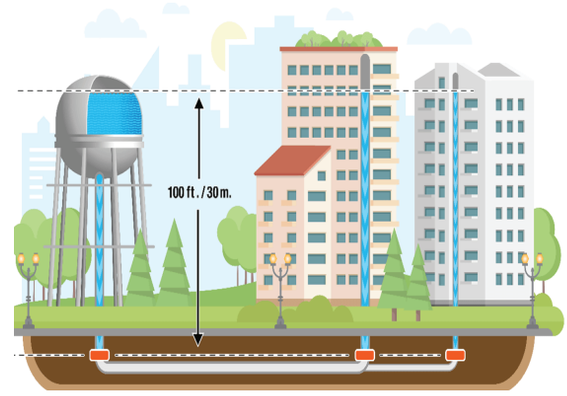
R_{eq} = _____ R_{eq} = _____



Chapter 4 PRESSURE

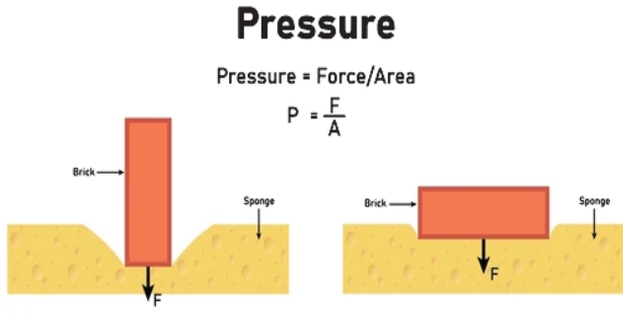
Objectives: After completing this unit you should be able to:

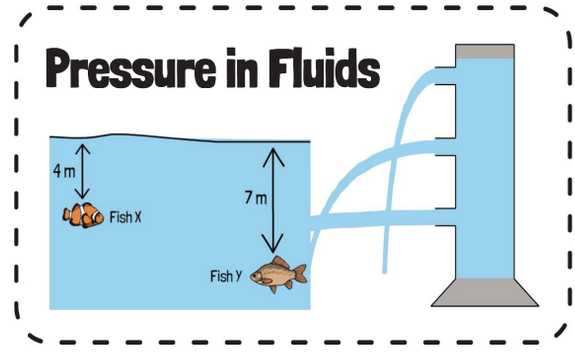
- ✓ understand concepts related to pressure;
- ✓ develop skill of manipulating problems related to pressure;
- ✓ appreciate the interrelatedness of all things;
- ✓ use a wide range of possibilities for developing knowledge of the major concepts with in physics.

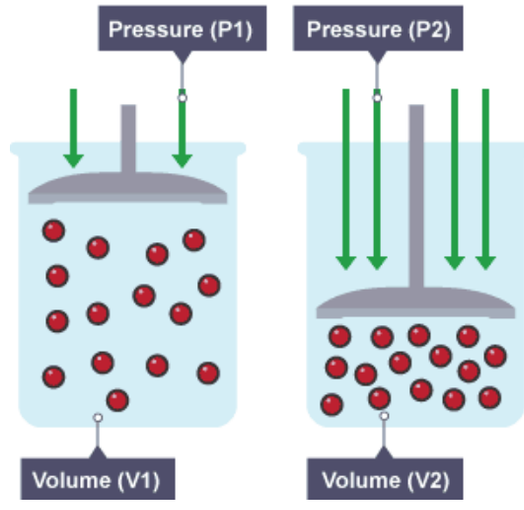


In this chapter you will study another effect of force known as **pressure**.

You begin first studying the concept of pressure using solid materials. Then you will learn about liquid pressure, atmospheric pressure, measuring atmospheric pressure, Pascal's principle and finally some applications of pressure.







Lesson 1 Pressure of Solids

1.Mass and Weight

Some terms that are commonly used in your everyday language have different meanings when used in Physics. The terms 'mass' and 'weight' have the same meaning in everyday language. However, in Physics, mass and weight are different. Let's look at how mass is defined.

The mass of a body is a measure of the amount of substance in the body.

The SI unit of mass is kilogram (kg).

Compare and contrast the definition of mass with weight:

The weight of a body is the force exerted on it due to the pull of gravity.

Since weight is a force, the SI unit of weight is newton (N).

Table 4.1 Differences between mass and weight

Mass	Weight
1.Remains constant	1.May change from place to place
2.is measured in kilogram	2.is measured in Newton
3.is not a force	3. is a force

Gravitational field Strength

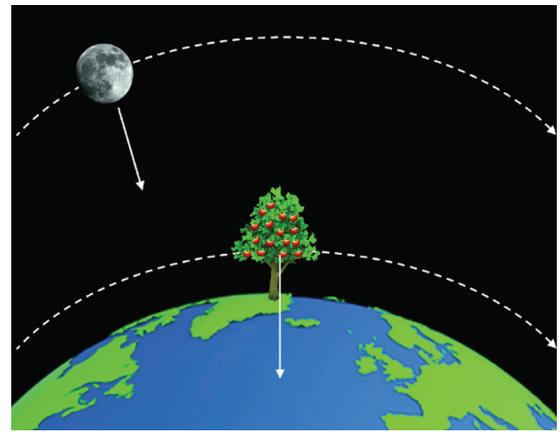


Fig. 1.The Earth's gravitational field

Figure 1 shows the Earth floating in space. There is a region around the Earth where you would experience an attractive force due to the Earth. As you move further away from the Earth, you will find that the force of attraction decreases. This region of influence of the Earth on you is called the gravitational field.

A gravitational field is a region in which a mass experiences a force due to gravity.

The strength of such a field measures how much gravitational force is experienced by a mass. It is called the gravitational field strength.
The SI unit of gravitational field strength is newton per kilogram (N/kg) The Earth's gravitational field strength has the symbol g and 9.81 N/kg. For calculation purposes, you can use g = 10 N/kg.
In the Earth's gravitational field, a mass of 1kg will experience a force or weight of 10 N. Therefore,
Weight =Mass x Gravitational field strength
In symbols, **W=mxg**
If you travel to the Moon where the gravitational field strength is 1.6 N/kg, you will find that your weight has changed.

2. Definition of Pressure

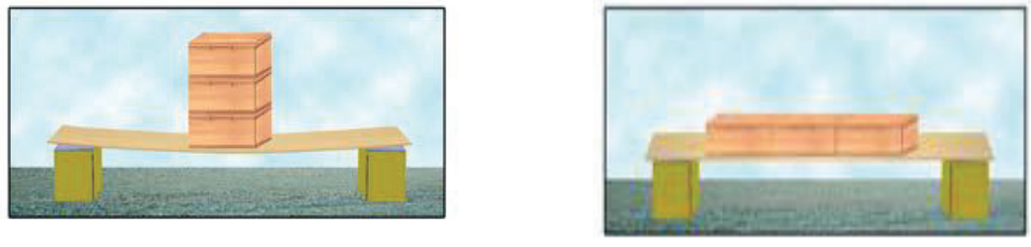
Activity 1

Discuss the following questions with your friends or parents.

1. **What is a force? What is an area?**
2. **Explain the effect of force on a unit area.**
3. **What do you call this effect?**

In Activity 1., most likely you recognized the difference between pressure and force.

Force is a push or a pull which changes the shape of a body; but pressure means force per unit area. Without changing the magnitude of the force you can have different pressures, just by changing the area on which the force acts. (see Fig 1).



a) Vertically arranged blocks ***b) Horizontally arranged blocks***

Fig 1. Effects of area on pressure

Activity 2. Observing the dependence of pressure on an area

Material required: blocks of wood (or books), a ruler, and a supporting body
Procedure: Place the ruler between two supporting bodies.

- Place the blocks of wood, one by one on the ruler. (Fig 1.) the blocks could be placed

a) Vertically (Fig. 1a) one over the other and the effect of the weight of the blocks on the ruler will be seen. What do you observe?
b) Horizontally (Fig. 1b) next to each other. Compare the effect of the weight of the blocks on the ruler. In which cases is the ruler bent more? Explain the reason.
- What can you conclude about the dependence of pressure on an area?

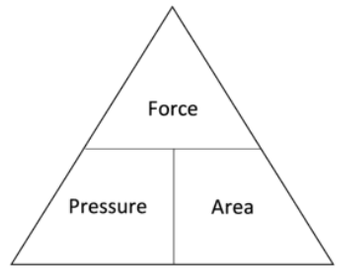
From Activity 2. you have noticed that, the pressure exerted by the block of wood on the ruler depends on the area on which the blocks were resting. The pressure is higher when the blocks are vertically placed than when they are arranged horizontally.

Pressure is an effect which occurs when a force is applied on a surface. Pressure is the amount of force acting on a unit area. The symbol of pressure is P.

$$\text{Pressure} = \frac{\text{Force exerted}}{\text{Area}} \Rightarrow P = \frac{F}{A}$$

The SI unit of pressure is Newton per square meter (N/m^2), and this unit is also called Pascal and is represented by Pa.

Where 1 Pa = $1N/m^2$.



Other non-SI units are the bar

1 millibar = 100Pa
1 bar = 100,000Pa

1.01 bar= 101,000 Pa and it is known as Standard Atmospheric Pressure

From the definition of pressure one can understand that the pressure of the system increases with an increase of a force.

From the Activity 2; you have observed that pressure depends on the area on which the force is applied.

Example: If your mass is 70 kg and the total area of the soles of your feet is $0.2\,m^2$, what pressure would you exert on the ground?

Solution: The force on the ground is your weight: $F = mg = 70 \times 9.8 = 686$ N (we take g, the acceleration due to gravity, to be approximately 9.8 ms^{-2}). Hence to two significant figures the pressure is

$$P = \frac{F}{A} = \frac{70 \times 9.8}{0.2} = 3,400\,\text{Pa}.$$

Practice: Find the pressure from a force of 100 N on an area of $0.25m^2$?

(a) 400 Pa (b) 25 Pa (c) 4,000 Pa (d) 2.5 Pa

Challenging questions

1. Do you know an elephant? What about its hoof? Is it large or small?
2. What would happen to the elephant if it had a small hoof similar to a goat?

Nature seems to be aware of the factors affecting pressure. The hoof of an elephant covers a wide area. When it walks, its large weight is spread over a wide area of its hoof. So, its pressure or force per unit area is less. This helps the elephant to walk over soft ground without sinking much into it.

Challenging question

Can you explain why tractors have tires of large area than an automobile?

Problem solving

1. Calculate the pressure exerted at a point while a nail tip of area $0.1cm^2$ is hammered into a wooden block with a force of 20N.

Given	Required	Solution
$F = 20N$ $A = 0.1cm^2 = 0.00001m^2$	$P = ?$	$P = {}^{F}/_{A} = \frac{20\ N}{0.00001m^2}$ $= 2,000,000\ N/m^2$ or 2,000,000 pa

2. What force is required to produce a pressure of 20,000 N/m^2 when the surface is $2m^2$

Given	Required	Solution
$P = 20,000\ {}^{N}/_{m^2}$ $A = 2m^2$	$F = ?$	$P = \frac{F}{A} \Rightarrow F = P \times A$ $= 20,000\ \frac{N}{m^2} \times 2m^2$ $\Rightarrow F = 40,000N$

3. A rectangular block of mass 50 kg lies with a side 2m by 1m on a table. What is the pressure exerted by its weight on the table? (Fig 2.)

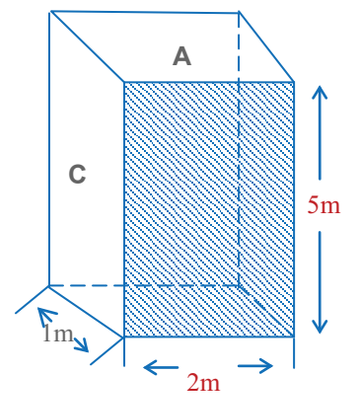


Fig 2. A block of sides 1m × 2m× 5m

Given	Required
Mass of block = 50 kg Sides of block = 2m×1m×5m g = 10 m/s^2	P_A = ?

Solution

From your discussion in unit one, you already learnt how to calculate the area of any surface. Hence area of the block on surface 'A' = 1m × 2m = 2m^2
Weight of the block W = m_b × g = 50kg × 10m/s^2 = 500N

The maximum pressure is exerted when the block is resting on the smallest surface, which is the minimum area. i.e. $P_A = \frac{F}{A} = \frac{W}{A} = \frac{500N}{2m^2}$ 250 N/m^2. On the contrary, when the surface area is maximum, the corresponding pressure will be minimum.

$P = \frac{F}{A} = \frac{W}{A}$ $P = \frac{500\ N}{10\ m^2}$ $P = 50\ N/m^2$

EXERCISES.

1.Complete the following table to find the force, area or pressure.

Force	Area	Pressure
400 N	2 m^2	
4 N	2 m^2	
80 kN		20 kPa
	3 m^2	6 kPa

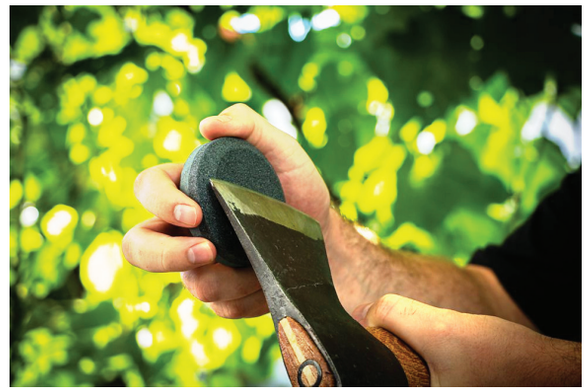
2.

(a) If a pressure gauge measures an increase in 3×10^4 Pa on an area of 0.07 m^2 what is the increase in the force applied to the area?
(b) Find the pressure produced by a kilogram of lead on a horizontal surface if the area it rests on is 0.02 m^2?

2. Application of Pressure of Solids

Understanding the connection between force, pressure and area can help scientists and engineers design and make more effective machines and devices.

For example, knives and cutting tools like axes have a small surface area, so they exert the largest possible pressure and cut things more easily. These tools often become blunt because the surface area of the cutting edge increases after repeated uses. Sharpening these tools ensures the surface area remains small, so the tools exert a larger pressure and will cut more effectively.



The opposite principle is used for designing skis and snowboards – these have a large surface area, to stop them from sinking into the snow.

The large surface area spreads the snowboarder's weight out, reducing the pressure exerted on the snow. This stops them from sinking, even if the snow is very deep.



Lesson 2 Pressure of Liquids

1. Liquid Pressure

To understand the nature of pressure in liquids, you need to remember some common characteristics of liquids.

Activity 3.

Discuss with your friends.

1. The differences and similarities between solid and liquid in terms of: (shape, volume, weight, density, etc)
2. Consider a plastic bag containing water. If it is pierced in different sides, what will happen to the water? (Fig 3.)

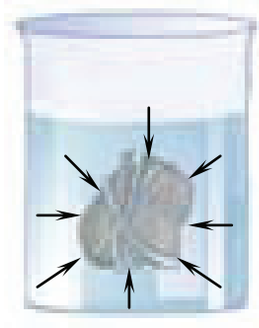
In your discussion in Activity 3, you might have talked that:

- Liquids have no definite shape; they take the shape of their containers.
- Liquids have definite volume, mass and weight.
- Compared to molecules of solids, the molecules of liquids are relatively free to move. The densities of solids are greater than densities of liquids.

This and some other activities show that liquids exert pressure on the walls of the container in all direction. As shown in Fig 3. when a plastic bag filled with water is pierced, then water flows in all directions.



a) Plastic bag filled with water.



b) Water pressures is the same in all directions on the object submerged

Fig 3 .Pressure in liquid acts in all directions

Activity 4

Containers of different shapes and sizes are half filled with the same liquid. They will rise to the same level.

i. What will be the pressure of the liquid at the bases of the containers?
ii. Is the pressure the same or different in different containers?
iii. Do the shapes of the containers affect the pressure?



Fig 4 Containers having different shapes filled with liquid to the same height.

Fig 5. illustrates the dependency of liquid pressure on height of the liquid in a container. The container has outlets 1, 2 and 3. The liquid coming out from outlet 3 travels longest distance while liquid coming from outlet 2 travels longer distance than outlet1.This shows that thepressureatoutlet3Fig.5.Pressure in liquids depends on the depth this greater than pressures at1 and 2.Activity 3.4 helps you to observe that,liquid pressure is independent of the shape ofthecontainer and the base area of the container. This means,the liquid pressure depends only ont he:

i) Depth (h) from the top surface of the liquid,
ii) Density of the liquid. The more denser the liquid the greater the pressure at any given depth.

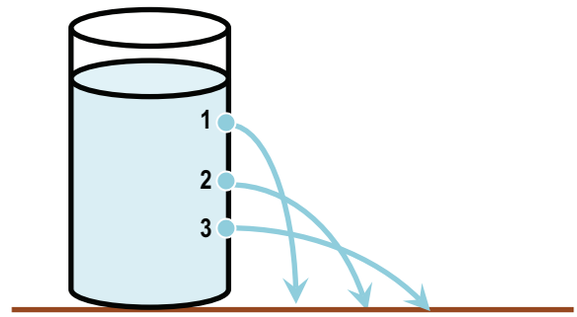
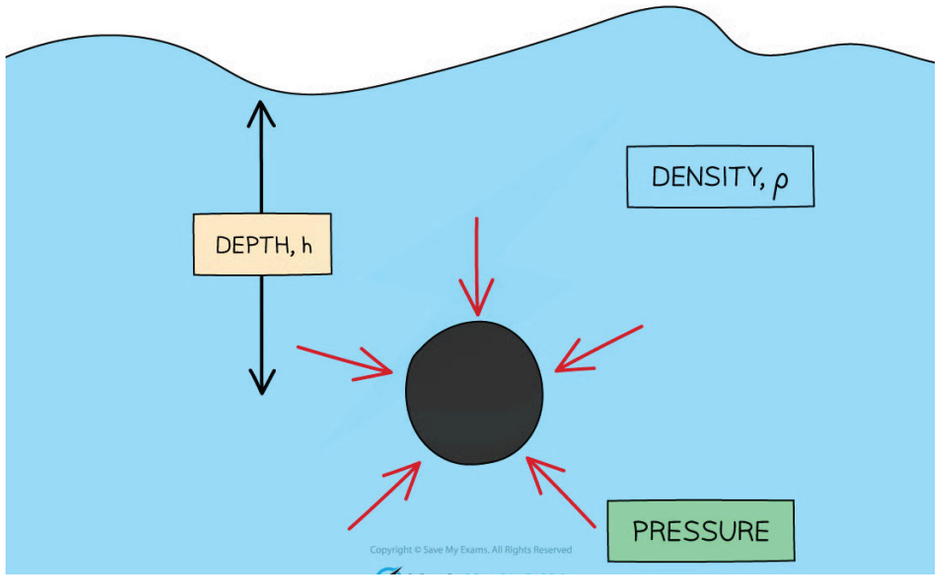


Fig.5. Pressure in liquids depends on the depth



Consider a liquid of density (ρ) in a container. At a certain depth (h), the force exerted by the liquid is equal to the weight of the liquid.

Thus the pressure, P = $\frac{F}{A} = \frac{mg}{A}$, where m is the mass.

But m= ρV, by substituting the two equations, you get

P = $\frac{F}{A} = \frac{mg}{A} = \frac{\rho Vg}{A}$,

Since the volume is the product of height and base area, (V = Ah) then

$$\Rightarrow P = \frac{\rho Ahg}{A} \quad \Rightarrow P = \rho gh$$

Consequently, the pressure exerted by a given liquid at a certain location is given by the equation P =ρgh;

Where p = pressure of liquid in N/m^2

ρ The density of liquid in kg/m^3

g = gravitational acceleration in m/s^2

h = depth of the liquid. (Height of the liquid above the point)

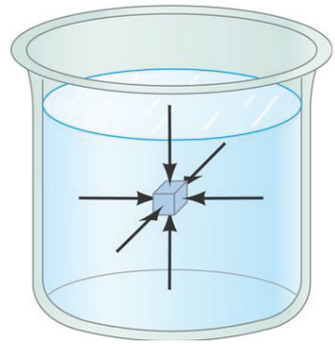
You can calculate the pressure exerted by water in a container, ocean, swimming pool, etc on an object, at any point (depth) of the water, using the equation of pressure, ($P = \rho gh$).

Activity 5.

Neglecting air pressure: find the pressure at a depth of 10m, 100m, 1000 m and at the bottom of the ocean (2000 m.) Consider density of ocean water 1030kg/m^3 and g = 10m/s^2.

Properties of pressure in a liquid at rest:

1. Pressure exists at every point within the liquid.
2. The pressure is the same at all points at the same level within a single liquid.
3. The pressure is directly proportional to the height (depth below the surface) and density of the liquid (ρ)
4. The force on the bottom of a container is the pressure at that level times the area of the bottom. (F = PA). This principle is used in the constructing of water dams.



Examples 2

1. A certain container is filled with water of density 1000kg/m^3 to a height of 50cm. How much pressure is exerted on the base of the container? (take g =10m/s^2).

Given	Required	Solution
$\rho = 1000kg/m^3$ h = 50cm = 0.5m g =10m/s^2	p = ?	P = ρgh P =1000kg/m^3×0.5m×10m/s^2 = 5000N/m^2 = 5000P$_a$

2. Water is poured into a vessel up to a height of 2m. (Calculate the pressure of water on the base of the vessel. (Take g = 10m/s^2, density of water ρ_ω = 1000kg/m^3).

Given	Required	Solution
h = 2m, g =10m/s^2 ρ_ω= 1000kg/m^3	P = ?	P = ρgh =1000kg/m^3 × 10m/s^2 × 2m = 2 × 10^{4}N/m^2 = 2× 10$^4 p_a$

Review Questions

1. Describe the Factors that affect liquid pressure.
2. Explain how liquid pressure varies with depth.

2. Pascal's Principle

Activity 6.

Can pressure be transmitted through a fluid? How?

A liquid can transmit any external pressure applied to it to all its parts. This property of liquid pressure is used in the hydraulic press, hydraulic brakes, to produce a large force from a small force.

A hydraulic press is consisted of a narrow cylinder connected to a wide cylinder both containing liquid (e.g. oil), and fitted with pistons A and B (as shown in Fig.6).

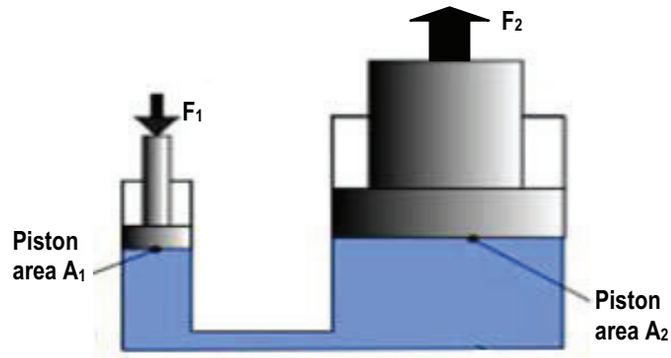


Fig 6.A- hydraulic press

Challenging question

Can you mention some other applications of hydraulic press principles?

Liquids and gases can flow and are called fluids. If an external pressure is applied to a confined fluid, the pressure will be transmitted to every point in the fluid. This principle is known as Pascal's principle.

Pascal's principle states that "a pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid."

Apply a force F_1 on the smaller piston of cross sectional area A_1, directly on the liquid such as oil. Then the pressure is $P_1 = \frac{F_1}{A_1}$.This pressure is transmitted undiminished through the oil to the larger cylinder, which has area of A_2. Since the pressure must be the same on both sides;

$P_1 = P_2$ where $P_2 = \frac{F_2}{A_2}$. The force exerted on piston A_2 is $P_2 A_2 = F_2$. The force F_2 is much greater than F_1. Thus, Pascal's principle is expressed mathematically as

$$P_1 = P_2$$
$$\Rightarrow \frac{F_1}{A_1} = \frac{F_2}{A_2}$$
$$\text{or } F_2 = \frac{A_2}{A_1} \times F_1$$

Pascal's principle is applied in technological devices which are used to produce great forces as in the case of hydraulic-press and brakes.

Consider Fig 6., which illustrates the working principle of a hydraulic press.

Examples 3.

1. In a hydraulic press the small piston has an area of 8 cm^2 while the large piston has an area of 400cm^2. If a force of 20N is applied on the small piston, what is the force on the larger piston?

Given	Required	Solution
$A_1 = 8cm^2$ $= 0.0008m^2$ $A_2 = 400cm^2$ $= 0.04\ m^2$ $F_1 = 20N$	$F_2 = ?$	$\frac{F_1}{A_1} = \frac{F_2}{A_2} \Rightarrow F_2 = \frac{A_2 \times F_1}{A_1}$ $= \frac{0.04m^2 \times 20N}{0.0008m^2}$ $= 1000N$

2. What should be the area of the large piston of a hydraulic press, if it is to lift a load of 20,000N when a force of 200N is applied on the small piston of area $1cm^2$?

Given	Required	Solution
$F_1 = 200N$ $F_2 = 20{,}000N$ $A_1 = 1cm^2$	$A_2 = ?$	$\frac{F_1}{A_1} = \frac{F_2}{A_2}$ $\Rightarrow A_2 = \frac{F_2}{F_1} \times A_1$ $= \frac{20{,}000N}{200N} \times 1$ $= 100cm^2$

3. In a hydraulic press the small piston has an area of $6cm^2$ while the large piston has an area of 300 cm^2. If a force of 30N is applied on the small piston, what load can it lift up (support)?

Given	Required	Solution
$A_1 = 6cm^2$ $A_2 = 300cm^2$ $F_1 = 30N$	$F_2 = ?$	$\frac{F_1}{A_1} = \frac{F_2}{A_2}$ $\Rightarrow F_2 = \frac{F_1}{A1} \times A_2$ $= \frac{30N}{6cm^2} \times 300cm^2$ $= 1500N$

Review Questions

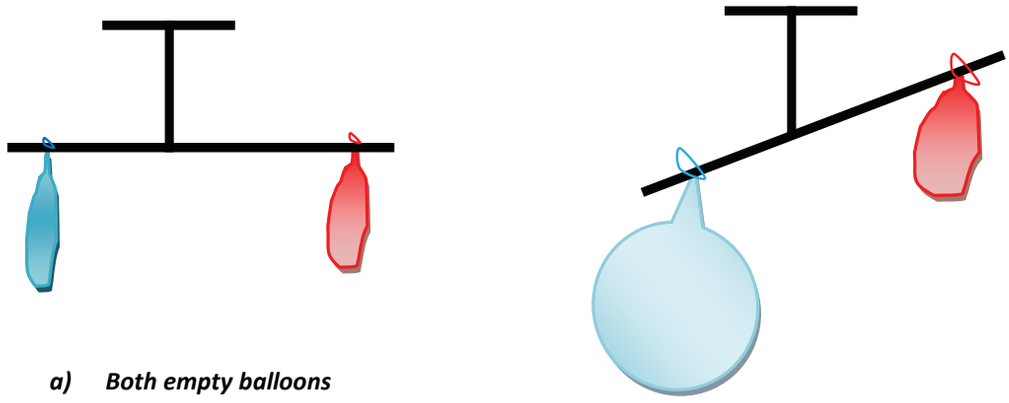
1. State Pascal's principle.
2. Draw a hydraulic press diagram and describe how it functions.

Lesson 3 Pressure of Gases

1. Atmospheric Pressure-Air Pressure

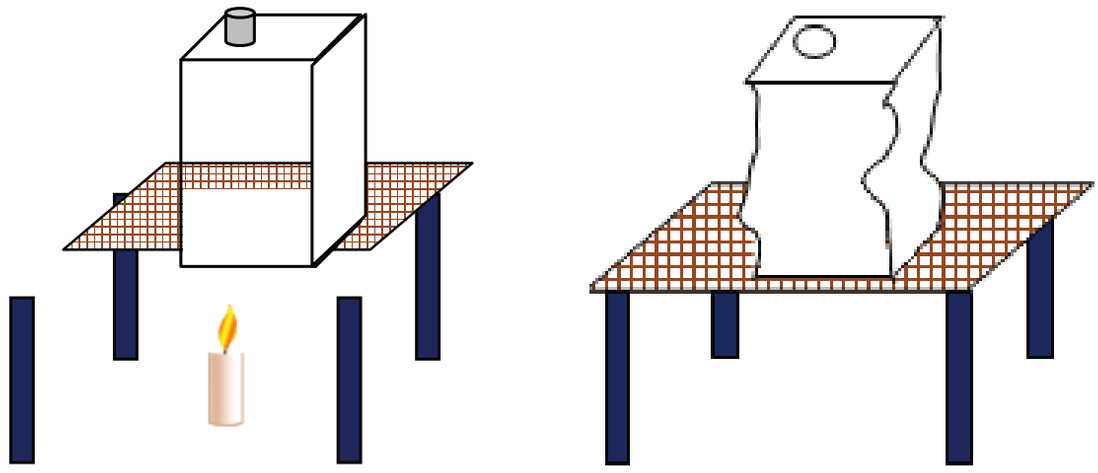
The earth is surrounded by a huge mass of air called the **Atmosphere**.

The force of gravity acts on air. Fig 7. illustrates that inflated balloon weighs more than empty balloon. Thus, air has a weight. The weight of atmosphere exerts pressure on bodies found on the earth's surface. This pressure is called **Atmospheric pressure**.



a) Both empty balloons *b) One balloon filled with air*

Fig 7. Air has a weight



a) Heating water in a can b) Water in a can is cooled by pouring cold water

Fig 8. Crushing can experiment

Activity 7 .Crushing can experiment

To observe the effect of atmospheric pressure. (Fig 8.)

Materials required: **Metal can, air tight stopper (cork), and small quantity of water, heat source (candle or Bunsen burner).**

Procedure **1. Pour a small quantity of water in the can and boil it for a few minutes until steam is driven out.**

2. Fit the cork tightly and remove the heat source away.

3.Pour cold water over the can so that the steam inside partly condenses into water and water vapor at low pressure.

Write a short report about your observation by answering the following questions;

1. What crushed the can?
2. What was the function of the heat and the steam inside the can?
3. What changed: mass, volume, density, temperature? Which quantity increased? decreased? Or remained the same?

From Activity 7. you see that, the can gets crushed inwards as in Fig.8 (b). This is due to the atmospheric pressure.

Activity 8. The U-tube experiment

Materials required: **U-tube that contains water, rubber tube.**

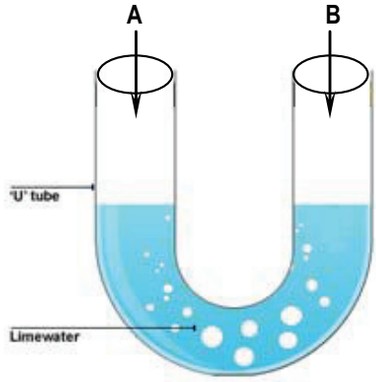
Procedure: **1. Pour water into the U-tube and mark the level.**

2. By connecting the rubber tube with the U-tube, blow air into it and mark the level of water you raised.

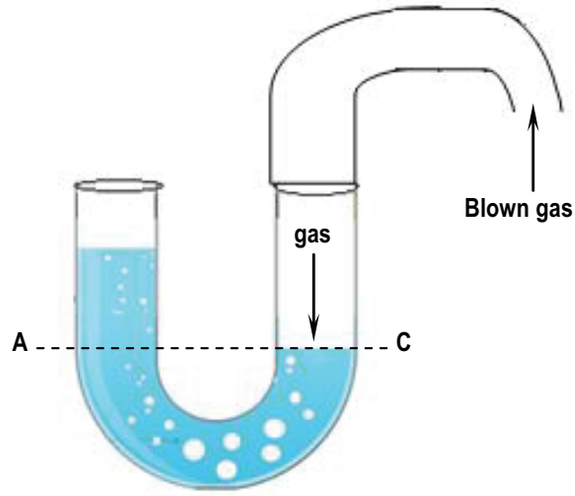
i. **What do you observe about the level of water in the two steps?**

ii. **Are the marks in the same level? Ask your friend to do the same and record the level of water raised.**

iii. **Explain what causes the level difference.**



a) Equal pressure keeps the water level equal



b) Extra pressure supports a column of liquid

Fig 9a,b. The U-tu be experiment

In Activity 8. the water levels were the same before you blow air into the U-tube (Fig.9(a)). This is because; the same atmospheric pressure was exerted on sides A and B. But when air is gently blown into one of the tubes, say tube B, the level of water changes on the other side until the pressure at C becomes equal to the blown air pressure. Thus the blown air pressure supports a column of liquid in the other tube.

Challenging question

Explain why water comes to your mouth when you suck water in a glass using drinking straw?

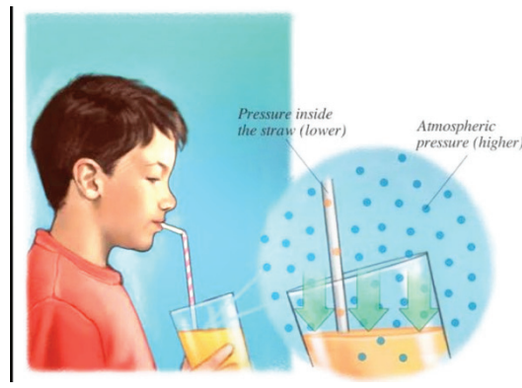


Fig 10. difference pressures makes drink flow.

Air pressure varies from place to place. If we travel up a mountain or go up in a hot air balloon, for example, the air pressure gets lesser and lesser as we go higher and higher.

Each layer of air above the earth presses down on the layers below, and so the greatest pressure is at the ground level, where we have the maximum amount of air above.

Atmospheric pressure is the force per unit area exerted on a surface by the weight of the atmosphere above that surface. It is caused by the weight of atmosphere above the measurement point. Low pressure areas have less atmospheric mass above their location, where as high pressure areas have more atmospheric mass above their location. Similarly, as elevation increases there is less overlying atmospheric mass, so that pressure decreases from sea level to the top of the atmosphere.

Challenging question

Explain how liquids are drawn into a syringe tube when the plunger is pulled out.

If we go up into the sky and reach the outer layer of the atmosphere called the stratosphere, the pressure will decrease until it reaches about zero. At this height there is hardly any air above it.

Check point 4.

1. Does the air around us exert pressure? Explain and give examples
2. What is atmospheric pressure?
3. What is the main purpose of doing "crushing can experiment? "
4. Explain how atmospheric pressure varies with the change of altitudes.

2. Measuring Atmospheric Pressure

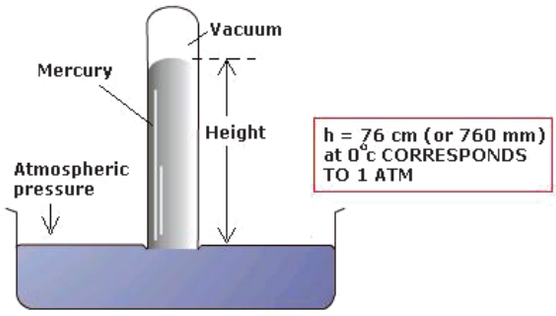
So far, you learnt the existence of atmospheric pressure. The next step is to ask how much strong this pressure is. To know the strength of atmospheric pressure, we need to measure it.

Atmospheric pressure is measured using an instrument called barometer. There are different kinds of barometers, but here we discuss only the mercury barometer.

Manometer is an instrument used to measure air pressure.

Challenging question

Explain the difference between air pressure and atmospheric pressure.



a) Structure of mercury barometer



Fig.11 Mercury Barometer **b) Actual Mercury barometer**

Mercury Barometer: A glass tube about 80 cm long is completely filled with clean dry mercury. It is then inverted in a basin of mercury under the surface of mercury (Fig. 3.10(a)). The level of mercury in the tube falls leaving an empty space (vacuum) at the top. This apparatus is called the mercury barometer.

The column of mercury in the tube is supported by the atmospheric pressure, P_A acting on the surface of the mercury in the open dish. The exact height of the mercury column depends on the pressure of the atmosphere. Therefore, atmospheric pressure is measured by the height of mercury it supports.

The atmospheric pressure at sea level supports 76 cm of mercury. This means that atmospheric pressure equals the pressure at the bottom of a column of mercury 76 cm high or 760 mm of mercury.

Sometimes atmospheric pressure is expressed in torrs, Where: 1 torr =1mm of Mercury (Hg)

To find the value of atmospheric pressure in the SI unit, we use $p = \rho gh$ where $g = 10 m/s^2$, $\rho = 1.36\ 10^4\ kg/m^3$, (the density of mercury) and $h = 0.76$ m,

Thus 76 cm of Hg equals $1 \times 10^5\ N/m^2 = 1 \times 10^5$ Pa.

×

Standard atmospheric pressure is taken to support a 760 mm high column of mercury at sea-level. That standard pressure is defined as the pressure at the foot of a column of mercury 760 mm high, equal to 1.01×10^5pa.=1 atmosphere. Another unit of pressure used in weather is the millibar (mb). 1mb=100Pa.

Relationships between different units of atmospheric pressure

1.01 bar = 101,000 Pa=760Torrs

Activity 9

1. **Explain why people don't use water barometer instead of mercury barometers?**
2. **Mention at least 3- disadvantages of using water in a barometer.**
3. **What are the major advantages of using mercury in a barometer?**

Suppose we use a water barometer at sea level where atmospheric pressure is 76 cm of Hg. This means the atmosphere supports 76 cm of a column of mercury. Now let us calculate the height (h_w) of the water barometer that the atmosphere can support.

Let h_m and h_w be heights of mercury and water supported by the atmospheric pressure respectively (see Fig 11 a and b).

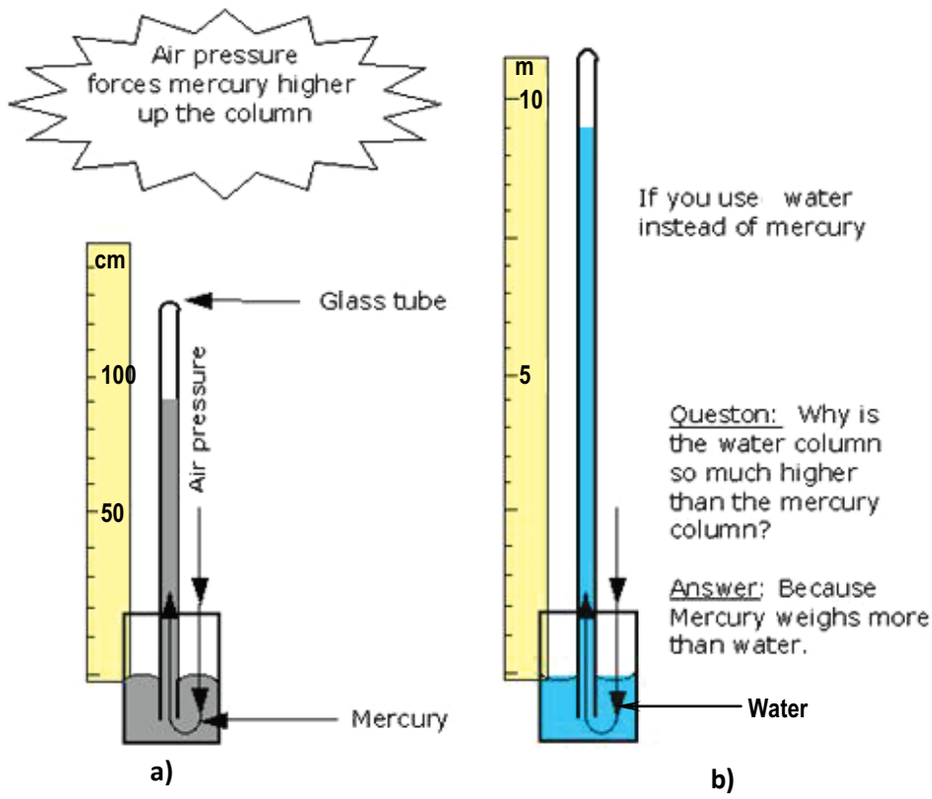


Fig 12 Comparison of mercury and water barometer

The atmosphere exerts the same pressure on the two barometers as they are placed at the same location. Hence the necessary height of water which is equivalent to 76 cm of Hg is 1034 cm or 10.34 m. That is, the water column rises up to 10.34 m through the tube. This means using water in a barometer requires us to have a tube taller than 10 m. Mercury barometers are small in size as the mercury rises only up to 76 cm.

Generally, mercury is selected for constructing a barometer due to the following advantages:

1. It is dense and therefore only a relatively short column of it is needed.
2. It is easy to see.
3. It doesn't freeze in very cold weather.
4. Very little is lost by evaporation.

Examples 3.4

1. What is the pressure in torr of a region whose atmospheric pressure is 64 cm-Hg?

Given	Required	Solution
P= 64 cm –Hg	P in 'torr' =?	Since 1cm =10mm, then P = 64 cm Hg = 640 mm-Hg. But 1mm Hg = 1 torr, hence 640mm- Hg = 640 torr

2. In a certain area the atmospheric pressure is 820 torr. Express this pressure in Pascal, bar and millibar.

Given	Required
P (in torr) = 820 torr	i) P(in Pascal)= ? ii) P (in bar) =? iii) P (in millibar) = ?

Solution

i) 760 torr = 101,000 pa
820 torr =?

$\Rightarrow$P(in pa) = $\frac{101{,}000\text{Pa} \times 820\cancel{\text{torr}}}{760\cancel{\text{torr}}}$

108,933.68Pa=

ii) If 760 torr = 1.01 bars
then 820 torr = ?

$\Rightarrow$ P (in bar) = $\frac{1.01\ \text{bar} \times 820\ \text{torr}}{760\ \text{torr}}$

= 1.08 bar

iii) Since 1bar
= 1000 millibar

$\Rightarrow$P (in millibar) =
1.08 × 1000 millibar
= 1080 millibar

Check point

1. Explain how atmospheric pressure is measured in terms of the columns of liquids it supports.
2. Discuss the units of atmospheric pressure.
3. What is a barometric reading for standard atmospheric pressure?

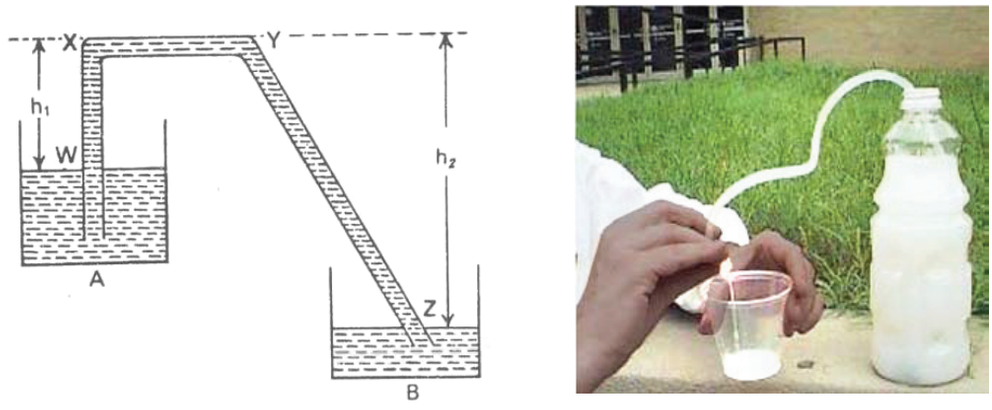
3.Applications of Atmospheric Pressure

There are many different devices that operate by the application of atmospheric pressure. Siphons, lift pumps, syringes and rubber-suckers are some technological gadgets that apply the atmospheric pressure.

3.1. Siphon

A siphon is one of the applications of the knowledge of atmospheric pressure.

A siphon is a device that transfers a liquid from a higher level to a lower level without disturbing the whole liquid or displacing the vessels.



a) b)

Fig 13 Siphons

"Siphon" is a tube or pipe which allows water (or another liquid) to flow from a high place, over a higher point and then down to a lower level. Siphons are

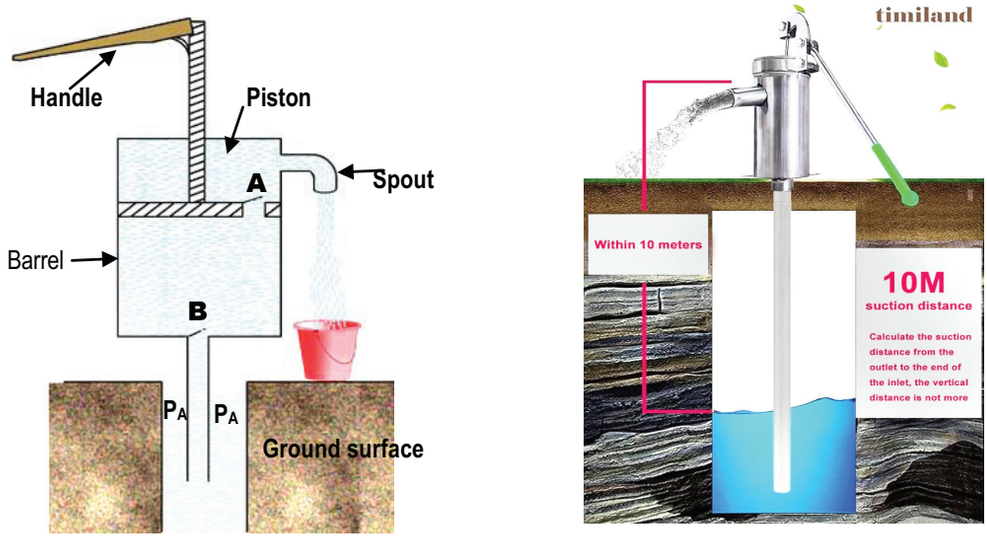
capable of raising water over a barrier. This is what makes them distinctive and interesting, as well as highly useful. A siphon is used in farm lands to move water from a reservoir to lower places.

It is also used to draw petroleum from a car tanker to a jerry can placed outside of a car. Try to observe such usages of siphon in your locality.

3.2. Lift pump

A lift pump is a widely used device in different regions of our country. You might have observed it in the television set, when the rural people are using it to pump water from deep well. It is very useful to our community as shown in Fig 14.

A lift pump has two valves, one on the piston and the other at the bottom of the barrel.



a) Structure of lift pump

b) The lift pump in action

Fig 14. Lift pump

To start the pump working, move the handle downwards so that valve 'B' closes and water inside the pump passes upward through valve A. When the handle moves up, valve A is closed and valve B is opened and water is pushed up the pipe through the valve 'B' by the atmospheric pressure. As the handle moves up and down continuously the atmospheric pressure pushes out the water through the spout.

3.3. The syringe

You are all familiar with a syringe from your early childhood. You used to play with it.

A syringe is used by doctors and nurses to give injection. It consists of a tightly fitted piston in a barrel.

The barrel will be filled by putting the nozzle under the liquid and drawing back the piston. This reduces the air pressure in the barrel and atmospheric pressure forces the liquid up into it. Pushing down the piston drives the liquid out of the nozzle. You can check how a syringe works by yourself.



Fig 15. The syringes

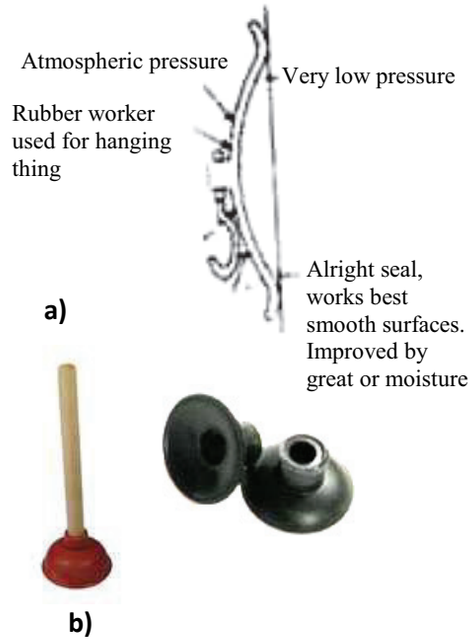


Fig 16. The Rubber sucker

3.4. The rubber sucker

A simple and a very useful application of atmospheric pressure is found in a circular shallow rubber-cup known as **sucker**. Rubber suckers are used for attaching pictures on shops and car windows (Fig 16a). You might have observed these suckers in the windows of a taxi where picture are hung when you come to school. When the sucker is moisture to obtain a good air seal and pressed on a smooth flat surface, the cup is flattened and air is squeeze out from the sucker. Then atmospheric pressure holds the sucker firmly on to the surface. Rubber suckers are also used in kitchen sinks to get rid of materials that block the flow of water. (Fig 16b).

3.5. Drinking straw

Drinking straw shown in Fig 17 works with the help of atmospheric pressure. When you suck the air in the straw into your mouth you create a vacuum in it. The atmospheric pressure on the surface of the juice in the glass forces the juice to flow into the straws. As you continue to suck the air, the juice reaches your month.

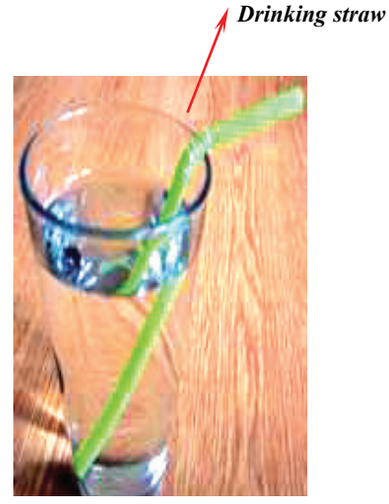


Fig 17. Drinking straw

Check point 3.6

1. Mention some practical applications of atmospheric pressure.
2. Explain how siphons, pumps, syringes, rubber suckers function.

SUMMARY

In this unit you learnt that:

- pressure is force per unit area P=F/A. Its SI unit is N/m^2 or Pascal Pa. One Pascal of pressure is a force of one Newton being exerted on an area of one square meter.
  i.e 1 pa = 1N/m^2
- pressure in solids depends on the area where the force is being exerts. Pressure in liquid at rest acts in all directions. Pressure in liquids depends on the depth of the liquid and the density of the liquid.(P=ρgh)
- Pascal's law states that "the pressure applied to a confined fluid is transmitted undiminished to every point of the fluid". Its application is the hydraulic press.
  i.e. $\frac{F_1}{A_1} = \frac{F_2}{A_2}$
- the air surrounding us has weight and exerts pressure on the surface of the earth. This pressure is called atmospheric pressure. A barometer is a device used to measure atmospheric pressure. The standard atmospheric pressure is 76 cm of Hg (mercury) or
  760 mm Hg = 1 atm; where 1 atm = 1.01$\times 10^5$ pa
- atmospheric Pressure plays a very important role in devices called pumps, siphon and rubber suckers. They operate themselves by the decrease and increase of pressure produced within the fluid.

Review Questions and Problems

I. Fill in the blank spaces with appropriate word or phrase.

1. **The SI unit of pressure is _____ .**
2. **The two main factors affecting pressure in a liquid are ___and ___.**
3. **The instrument used to measure atmospheric pressure is _____.**
4. **Standard atmospheric pressure is equal to ____ Pa or _____cm of Hg.**

II. Short answer Questions

1. **What is a pressure?**
2. **On what factors do pressure depend?**
3. **What mathematical expression do you use to calculate the pressure due to liquid at a given depth? Explain each term.**
4. **Describe the working principle of a hydraulic press using Pascal's principle.**
5. **List some applications of atmospheric pressure.**
6. **Draw a lift pump diagram and explain how it operates.**

III. Solve the following problems

1. **A metal box of weigh 20 N rests on its 1 m by 0.6m side on floor. How much is the pressure exerted by the metal box on the floor? Take g= 10 m/s^2.**
2. **A rectangular container of base 50 cm by 30 cm is filled with water to a depth of 5 cm. How much is the pressure exerted at the base?**
   (Take ρ_w= 1000 kg/m^3 and g =10 m/s^2)
3. **How much is the pressure exerted at a point where a ball- pointed –pen of area 1mm^2 is pushed against the paper with a force of 24 N?**
4. **Suppose a scientist was able to construct a barometer with a liquid being twice denser than mercury, then how high would the liquid raise at standard pressure?**
5. **A cylindrical container is filled with oil to a depth of 32 cm. if the pressure exerted at the base of the container is known to be 2560pa then, how much is the density of the oil?**

Glossary

Acceleration: Rate of change of velocity.
Acceleration of gravity: Rate of change of velocity due to gravitational attraction of the earth.
Ammeter: Electric device used to measure current.
Ampere: Unit for the rate of flow of charged particles. One ampere equals a flow of one coulomb of charge per second.
Amplitude: Maximum displacement from zero of any periodic phenomenon.
Aneroid barometer: Device using a partially evacuated box to measure atmospheric pressure.
Angle of incidence: Angle between a light ray and a line perpendicular to the surface that the ray is striking.
Angle of refraction: Angle between a refracted ray and a line perpendicular to the surface that the ray is leaving.
Antenna: Device used to receive or radiate electromagnetic waves.
Archimedes' principle: An object immerses in a fluid is buoyed up by a force equal to the weight of the fluid displaced by the object.
Armature: Coil of wire that produces current in a generator and rotation in an electric motor.
Barometer: Device for measuring the pressure of the atmosphere.
Battery or galvanic cell: Two dissimilar conductors and an electrolyte that produces a potential difference.
Bernoulli's principle: The pressure exerted by a fluid decreases as its velocity increases.
Black hole: Collapsed astronomical object of sufficient mass to prevent the escape of light.
Components of a vector: Two or more vectors (usually perpendicular) that when added together, produce the original vector.
Concurrent forces: Forces acting on the same point.
: Material through which charged particles move readily.
Control rods: Devices in a nuclear reactor used to regulate the rate of the nuclear reaction.
Convection current: Current caused by motion of a body of fluid due to differences in density resulting from thermal expansion.
Converging lens: Lens, thick in the middle and thin at the edge, that causes parallel rays to converge. **Converging mirror**: Concave mirror capable of causing parallel rays to converge.
Coulomb: Unit of quantity of electric charge equal to the charge found on electrons.
Candela: Unit of luminous intensity.
Center of curvature: Center of the sphere from which a spherical mirror is taken.
Centripetal acceleration: Acceleration always at right angles to the velocity of a particle.
Centripetal force: Force directed toward the center of a circle, keeps particles moving in uniform circular motion.
Critical angle: Minimum angle of incidence that produces total internal reflection.
Deceleration: Negative acceleration.
Diffraction: Bending of light waves around an object in its path.
Diffuse reflection: Reflected light scattered in many directions.
Displacement: Vector quantity representing the change in position of an object.
Distance: Scalar quantity equal to the sum of the magnitude of the displacements.
Diverging lens: Lens thin in the middle and thick at the edge, that causes parallel rays to diverge.
Diverging mirror: Convex mirror capable of causing parallel light rays to diverge.
Dynamics: Study of the motion of particles acted upon by forces.
Effective resistance: Resistance of a single resistor that could replace a combination of resistors.
Efficiency: Ratio of output work to input work.
Elasticity: Ability of object to return to its original form after removal of deforming forces.
Electric circuit: Continuous path that can be followed by charged particles.
Electric current: Flow of charged particles.

Electric field: The property of space around a charged object that causes forces on other charged objects.
Electric field intensity: Ratio of the force exerted by a field on a charged particle and the charge on the particles.
Electric field lines: Lines representing the direction of the electric field.
Electric force: Force between two objects due to their charges.
Electromagnet: Device in which a magnetic field is generated by an electric current.
Electromagnetic force: One of the fundamental forces. The force that exists between electric charges.
Electromagnetic induction: Generation of an electric current by having a wire cut (or cut by) magnetic flux lines.
Electromagnetic wave: Wave consisting of electric and magnetic fields that move at the speed of light in space.
Electromagnetism: Interrelationship of magnetic fields and electric currents.
Electromotive force (EMF): Potential difference generated by electromagnetic induction. **Energy**: Capacity to do work.
Equilibrant: Force equal in magnitude to a resultant, but opposite in direction.
Equilibrium: Condition in which a net force on an object is zero.
Focal length: Distance from the focal point to the vertex of a mirror or lens.
Focal point: Point of convergence, real or apparent (virtual), of rays reflected by a mirror or refracted by a lens.
Force: Action that results in accelerating or deforming an object.
Frequency: Number of occurrences in a unit of time.
Friction: Force opposing motion between two objects that are in contact.
Generator: Device using mechanical energy to produce electric energy.
Gravitational field: Distortion of space due to the presence of a mass. See field.
Gravitational force: Attraction between objects due to their masses.
Grounding: Connecting a charged object to the earth to remove the object's charge.
Heat: Quantity of thermal energy transferred from one object to another object because of a difference in temperature.
Hertz: Unit of frequency equal to one event (cycle) per second.

Index of refraction: $n = \frac{\sin i}{\sin r}$, where n=index of refraction, i=angle of incidence, and r=angle of refraction.
Insulator: Material through which the flow of charged particles is greatly restricted.
Joule: Unit of work or energy equal to a Newton-meter.
Kilowatt hour: Amount of energy equal to $36 \times 10^6 J$.
Kinematics: Study of the motion of the particles.
Kinetic energy: Energy of an object due to its motion.
Law of conservation of energy: In non nuclear changes, energy can be neither created nor destroyed. **Length**: Distance between two points.
Lenz's law: The magnetic field generated by an induced current always opposes the field generating the current.
Light: Electromagnetic radiation of 4×10^{-7} meter to 7×10^{-7} meter wavelength.
Liquid: State of matter in which particles are in close proximity but may readily change their relative positions.
Longitudinal wave: Wave in which the disturbance is in the sane direction as the direction of travel of the wave.
Magnetic field: Space around a magnet in which magnetic force can be detected.
Magnetic force: Force between two object due to the magnetic flux of one or both objects.
Mechanical advantage (MA): Ratio of resistance force to effort force in a machine.

Mechanical wave: Disturbance traveling through a medium.
Normal force: Force perpendicular to a surface.

Ohm: Unit of electric resistance.
Ohm's law: Current flowing in a conductor varies directly with the potential difference and inversely with the resistance.
Opaque material: Material that does not transmit light.
Parallel circuit: Circuit in which there are two or more paths for the charged particles to follow as they complete the circuit.
Pascal: Unit of pressure equal to a Newton per square meter.
Period: Time duration of a phenomenon or event.
Potential energy: Energy of an object due to its position.
Power: Rate of doing work.
Pressure: Force per unit area.
Principle axis: Radius connecting the center of curvature of a spherical mirror with its geometric vertex.
Real image: Image formed by rays that actually recombine to form the image.
Receiver: Device used to detect electromagnetic waves.
Refraction: Change in direction of a wave front as it passes from one medium to another.
Resistance: Opposition to flow of electric current.
Resolution: To find the component of a vector in a given direction.
Resultant: Single force that has the same effect as two or more concurrent vectors.
Satellite: Object in orbit around a planet.
Scalar quantity: Quantity having magnitude (size) only.
Schematic diagram: Diagram of an electric circuit using symbols.
Series circuit: Circuit in which the charged particles must flow through each component of the circuit, one after the other.
SI: An internationally agreed-upon consistent method of using the metric system of measurement.
Significant digits: The reliable digits reported in a measurement.
Simple harmonic motion: Motion in which a particle repeats the same path periodically.
Simple machine: Lever, pulley, wheel and axle, inclined plane, wedge, or screw.
Snell's law: The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for two specific substances in contact where a light passes. See index of refraction.
Specific heat: Energy required to change the temperature of 1 kg of a substance I kelvin.
Spectrum: Array of the various wavelengths composing light.
Speed: Ratio of distance traveled to the time interval.
Speed of light: $300 \times 10^8 m/s$ (in vacuum).
Spherical concave mirror: A converging mirror that is formed of a spherical segment of one base.
Temperature: Measure of the average kinetic energy of molecules.
Terminal velocity: Velocity of a falling object when the flow of air resistance is equal to the weight.
Thermal energy: Sum of potential energy and kinetic energy or random motion of particles in an object.
Thermal excitation: Exciting an atom by heating it.
Thermal expansion: Moving apart of particles as their temperature rises.
Thermometer: Device used to measure temperature.
Transformer: Device used to transfer energy from one circuit to another circuit by mutual inductance across two coils.
Vector quantity: Quantity having both magnitude (size) and direction.
Velocity: Rate of change of position.
Virtual image: Image whose rays appear to emanate from a point without actually doing so. **Volts**: Unit of potential difference.
Voltmeter: Electric device used to measure potential difference.

Watt: Unit of power equal to 1 J/s.
Wave: Traveling disturbance in a field or medium.
Wavelength: Distance between corresponding points on two successive waves.
Work: Product of force and displacement.
Work function: Energy needed to remove an electron from metal as in the photoelectric effect. **X rays**: Electromagnetic waves of very short wavelength.